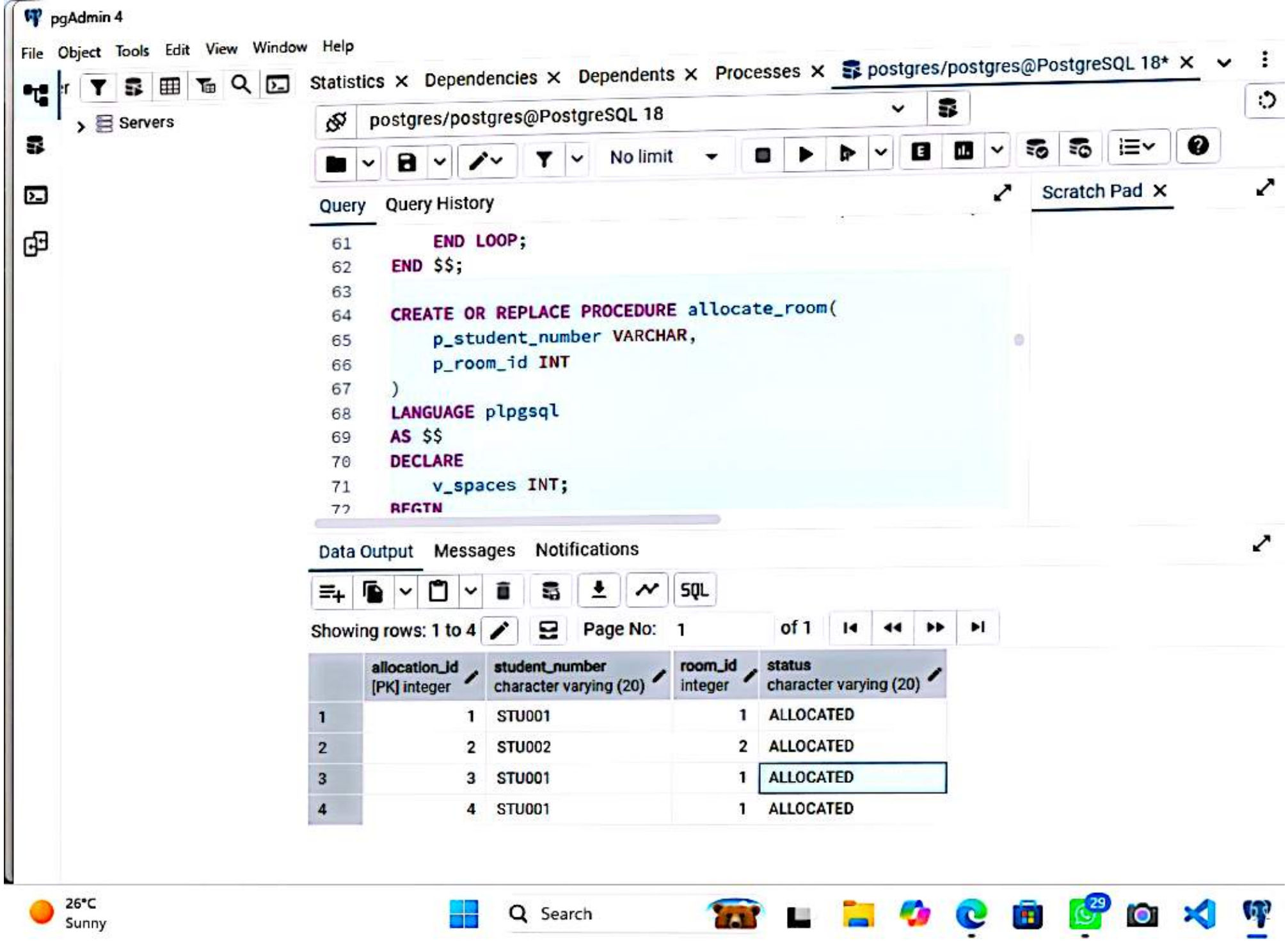

pgAdmin 4
File Object Tools Edit View Window Help
Statistics Dependencies Dependents Processes postgres/postgres@PostgreSQL 18*
Servers
postgres/postgres@PostgreSQL 18
No limit
Query Query History
Scratch Pad
61 END LOOP;
62 END $$;
63
64 CREATE OR REPLACE PROCEDURE allocate_room(
65 p_student_number VARCHAR,
66 p_room_id INT
67)
68 LANGUAGE plpgsql
69 AS $$
70 DECLARE
71 v_spaces INT;
72 BEGIN
Data Output Messages Notifications
SQL
Showing rows: 1 to 4 Page No: 1 of 1
allocation_id [PK] integer
student_number character varying (20)
room_id integer
status character varying (20)
1 1 STU001 1 ALLOCATED
2 2 STU002 2 ALLOCATED
3 3 STU001 1 ALLOCATED
4 4 STU001 1 ALLOCATED
26°C Sunny
Search

CamScanner

Help

Statistics × Dependencies × Dependents × Processes × postgres/postgres@PostgreSQL 18* ×

postgres/postgres@PostgreSQL 18

No limit

Query Query History

Execute script F5

Scratch Pad ×

```
59      FOR room_check IN 1..3 LOOP
60          RAISE NOTICE 'Room check number %', room_check;
61      END LOOP;
62  END $$;
63
64  CREATE OR REPLACE PROCEDURE allocate_room(
65      p_student_number VARCHAR,
66      p_room_id INT
67  )
68  LANGUAGE plpgsql
69  AS $$
70  DECLARE
```

Data Output Messages Notifications

SQL

Showing rows: 1 to 4 Page No: 1 of 1

	allocation_id [PK] integer	student_number character varying (20)	room_id integer	status character varying (20)
1	1	STU001	1	ALLOCATED
2	2	STU002	2	ALLOCATED
3	3	STU001	1	ALLOCATED
4	4	STU001	1	ALLOCATED

Search

CamScanner

Help

Statistics × Dependencies × Dependents × Processes × postgres/postgres@PostgreSQL 18* ×

postgres/postgres@PostgreSQL 18

No limit

Query Query History

Execute script F5

Scratch Pad ×

```
59      FOR room_check IN 1..3 LOOP
60          RAISE NOTICE 'Room check number %', room_check;
61      END LOOP;
62  END $$;
63
64  CREATE OR REPLACE PROCEDURE allocate_room(
65      p_student_number VARCHAR,
66      p_room_id INT
67  )
68  LANGUAGE plpgsql
69  AS $$
```

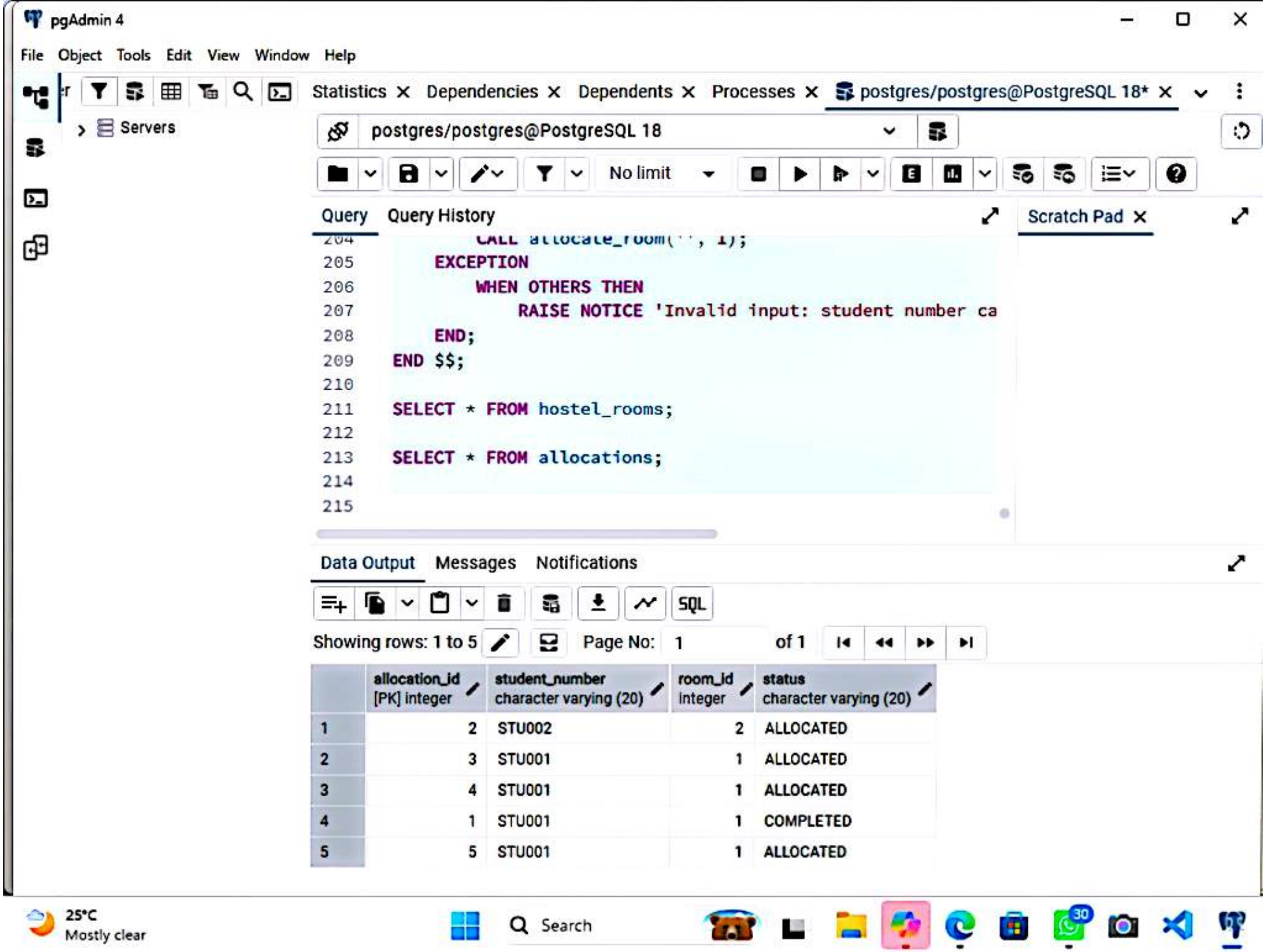
Data Output Messages Notifications

```
NOTICE:  Student STU001 allocated to room 1
NOTICE:  Room 2 is full. Allocation not recorded.
NOTICE:  Room 3 is full. Allocation not recorded.
NOTICE:  This allocation has already been completed. Bed space was NOT restored again.
NOTICE:  This allocation has already been completed. Bed space was NOT restored again.
CALL

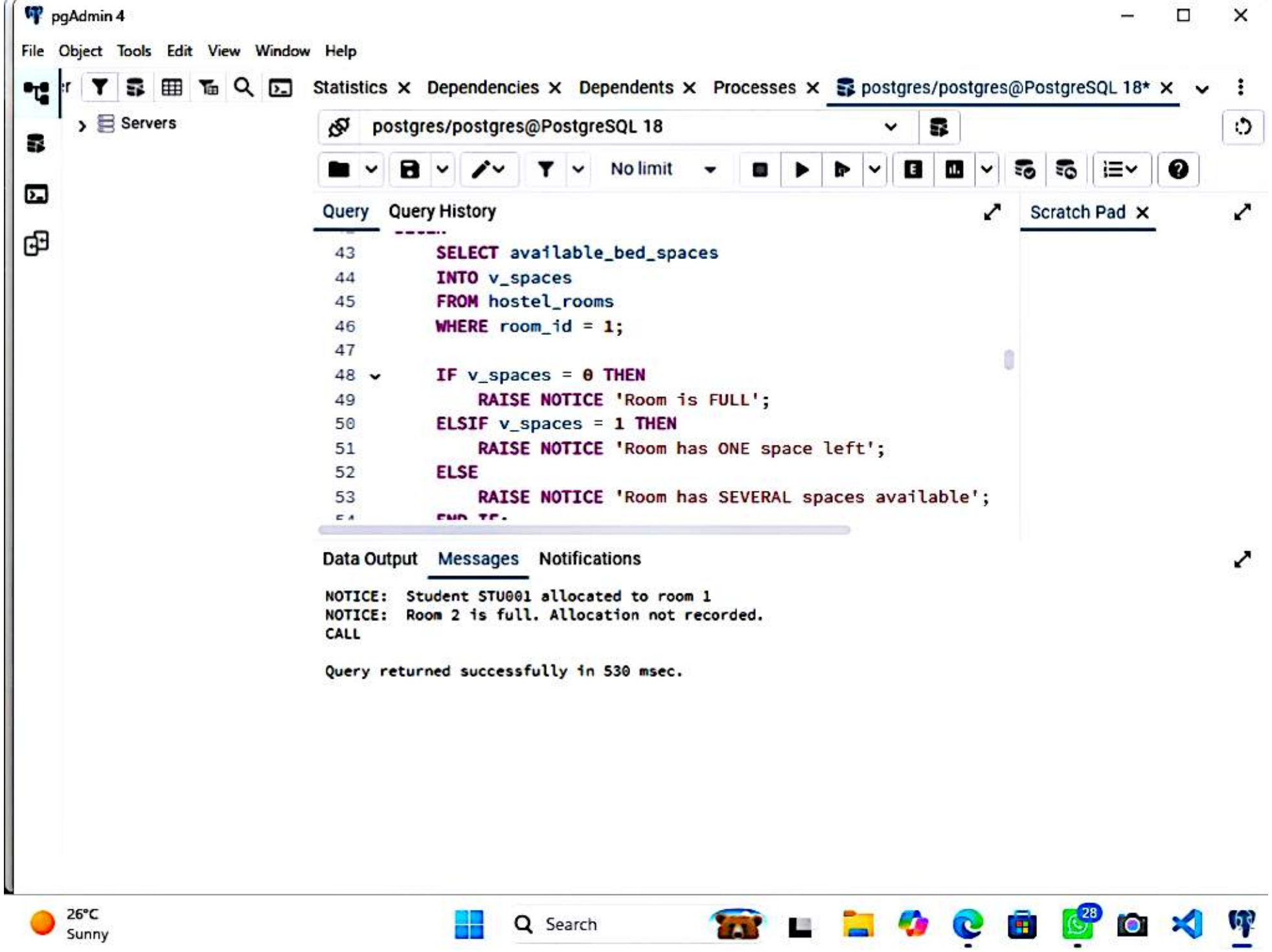
Query returned successfully in 432 msec.
```

Search

CamScanner


pgAdmin 4
File Object Tools Edit View Window Help
Statistics
Dependencies
Dependents
Processes
postgres/postgres@PostgreSQL 18*
Servers
postgres/postgres@PostgreSQL 18
No limit
Query
Query History
Scratch Pad
CALL allocate_room('', 1);
EXCEPTION
WHEN OTHERS THEN
RAISE NOTICE 'Invalid input: student number ca
END;
END $$;
SELECT * FROM hostel_rooms;
SELECT * FROM allocations;
Data Output
Messages
Notifications
SQL
Showing rows: 1 to 5
Page No: 1 of 1
allocation_id [PK] integer
student_number character varying (20)
room_id integer
status character varying (20)
1 2 STU002 2 ALLOCATED
2 3 STU001 1 ALLOCATED
3 4 STU001 1 ALLOCATED
4 1 STU001 1 COMPLETED
5 5 STU001 1 ALLOCATED
25°C
Mostly clear
Search

CamScanner


pgAdmin 4
File Object Tools Edit View Window Help
Statistics
Dependencies
Dependents
Processes
postgres/postgres@PostgreSQL 18*
Servers
postgres/postgres@PostgreSQL 18
No limit
Query
Query History
Scratch Pad
43 SELECT available_bed_spaces
44 INTO v_spaces
45 FROM hostel_rooms
46 WHERE room_id = 1;
47
48 IF v_spaces = 0 THEN
49 RAISE NOTICE 'Room is FULL';
50 ELSIF v_spaces = 1 THEN
51 RAISE NOTICE 'Room has ONE space left';
52 ELSE
53 RAISE NOTICE 'Room has SEVERAL spaces available';
Data Output
Messages
Notifications
NOTICE: Student STU001 allocated to room 1
NOTICE: Room 2 is full. Allocation not recorded.
CALL
Query returned successfully in 530 msec.
26°C
Sunny
Search

CamScanner

Help

Statistics × Dependencies × Dependents × Processes × postgres/postgres@PostgreSQL 18* ×

postgres/postgres@PostgreSQL 18

No limit

Query Query History

Scratch Pad ×

```
58  BEGIN
59      FOR room_check IN 1..3 LOOP
60          RAISE NOTICE 'Room check number %', room_check;
61      END LOOP;
62  END $$;
63
64  CREATE OR REPLACE PROCEDURE allocate_room(
65      p_student_number VARCHAR,
66      p_room_id INT
67  )
68  LANGUAGE plpgsql
69  AS $$
```

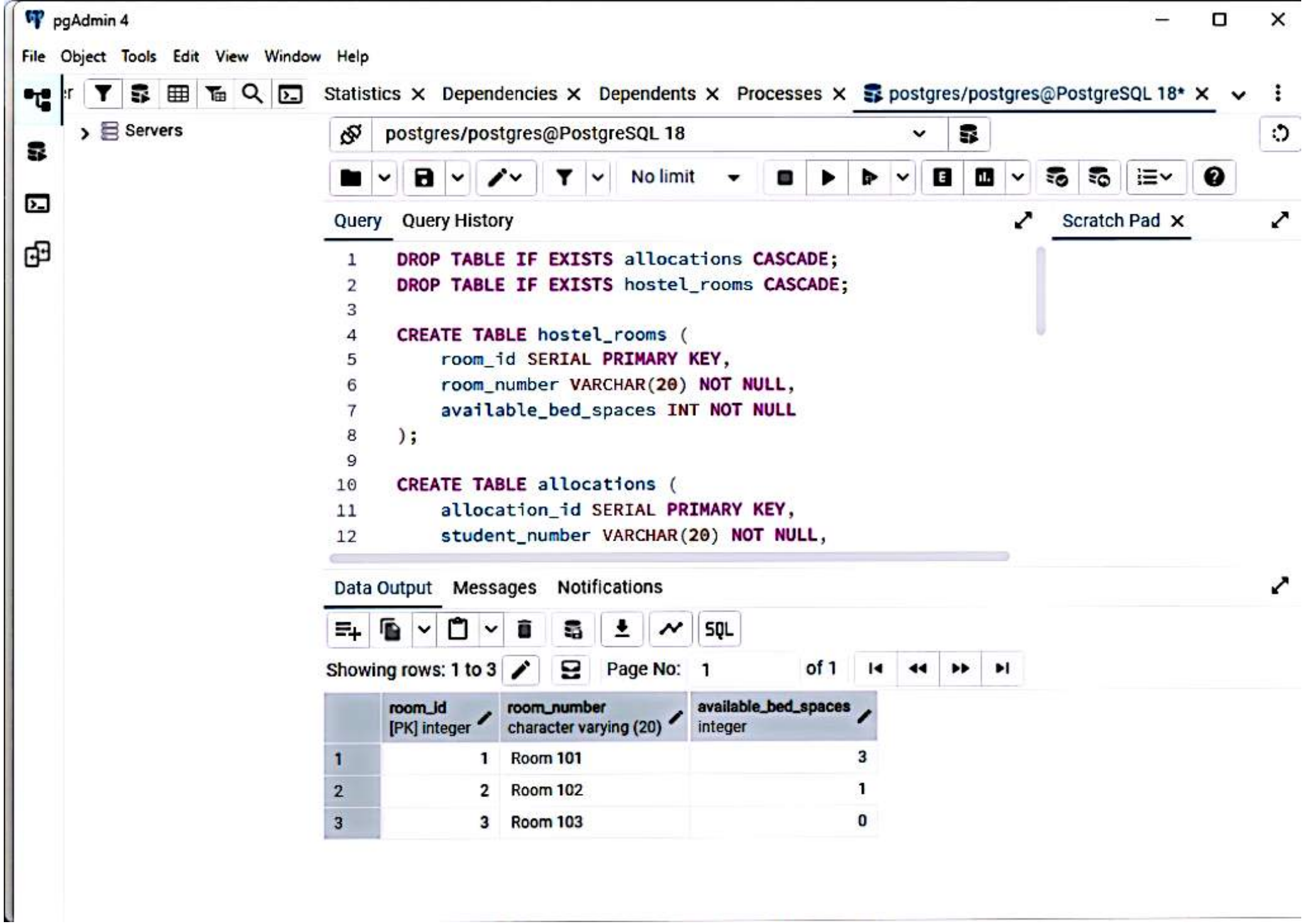
Data Output Messages Notifications

```
NOTICE:  Student STU001 allocated to room 1
NOTICE:  Room 2 is full. Allocation not recorded.
NOTICE:  Room 3 is full. Allocation not recorded.
CALL

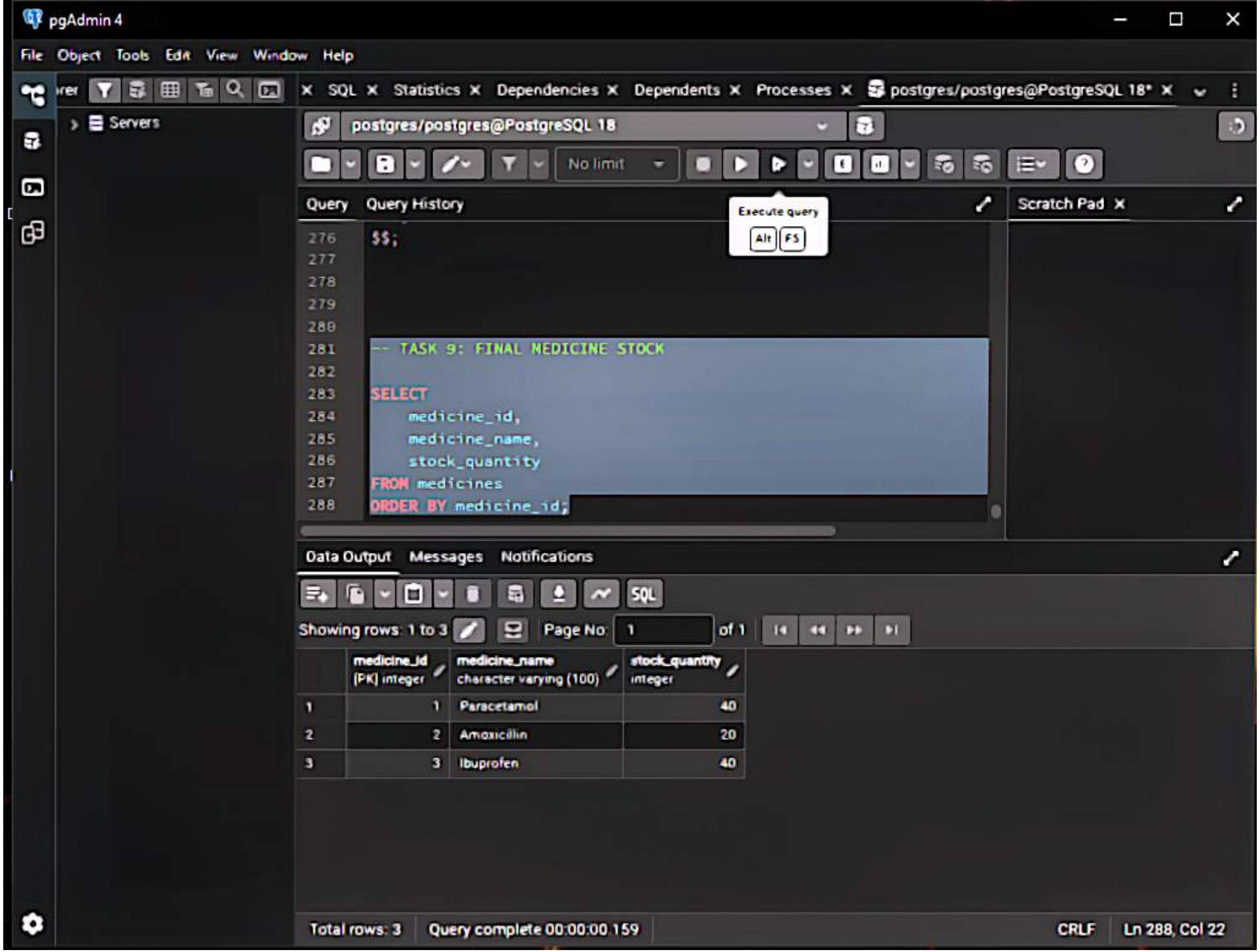
Query returned successfully in 485 msec.
```

Search

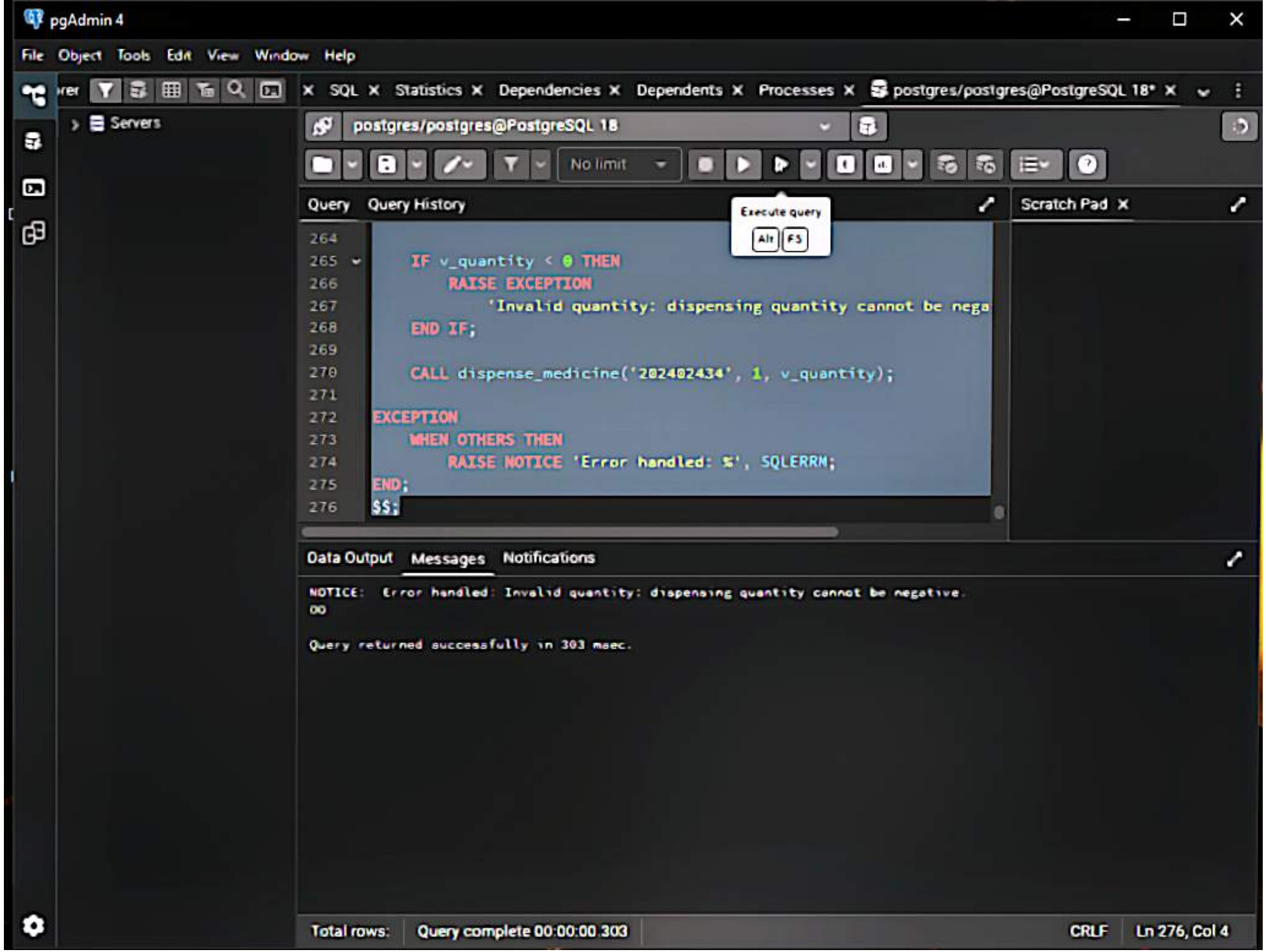
CamScanner


pgAdmin 4
File Object Tools Edit View Window Help
Statistics
Dependencies
Dependents
Processes
postgres/postgres@PostgreSQL 18*
Servers
postgres/postgres@PostgreSQL 18
No limit
Query
Query History
Scratch Pad
1 DROP TABLE IF EXISTS allocations CASCADE;
2 DROP TABLE IF EXISTS hostel_rooms CASCADE;
3
4 CREATE TABLE hostel_rooms (
5 room_id SERIAL PRIMARY KEY,
6 room_number VARCHAR(20) NOT NULL,
7 available_bed_spaces INT NOT NULL
8);
9
10 CREATE TABLE allocations (
11 allocation_id SERIAL PRIMARY KEY,
12 student_number VARCHAR(20) NOT NULL,
Data Output
Messages
Notifications
SQL
Showing rows: 1 to 3
Page No: 1 of 1
room_id [PK] integer
room_number character varying (20)
available_bed_spaces integer
1 1 Room 101 3
2 2 Room 102 1
3 3 Room 103 0

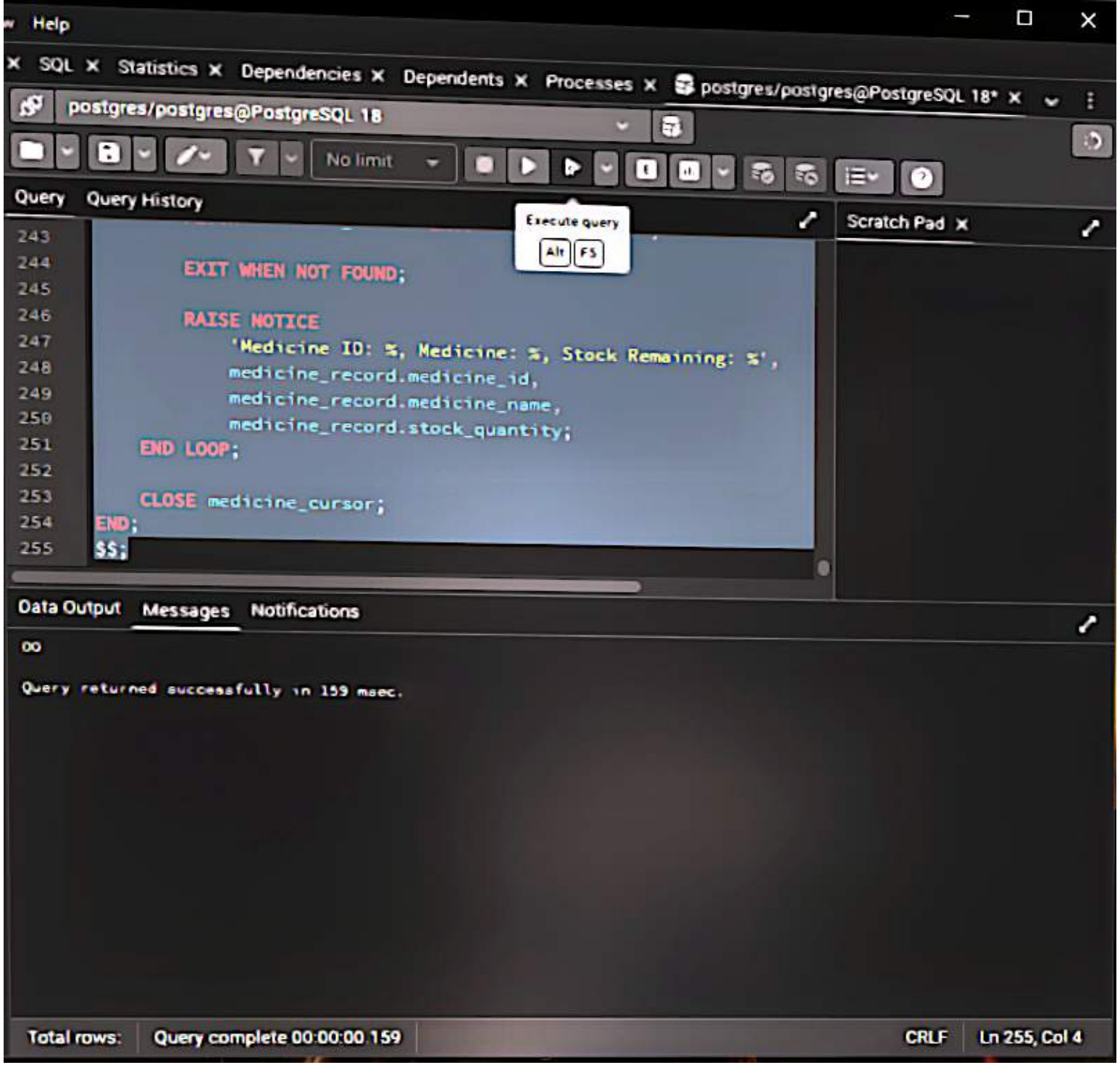
CamScanner


pgAdmin 4
File Object Tools Edit View Window Help
SQL Statistics Dependencies Dependents Processes postgres/postgres@PostgreSQL 18*
Servers
postgres/postgres@PostgreSQL 18
No limit
Query Query History
Execute query
Alt F5
Scratch Pad
276 $$;
277
278
279
280
281 -- TASK 9: FINAL MEDICINE STOCK
282
283 SELECT
284 medicine_id,
285 medicine_name,
286 stock_quantity
287 FROM medicines
288 ORDER BY medicine_id;
Data Output Messages Notifications
SQL
Showing rows: 1 to 3 Page No: 1 of 1
medicine_id [PK] integer
medicine_name character varying (100)
stock_quantity integer
1 1 Paracetamol 40
2 2 Amoxicillin 20
3 3 Ibuprofen 40
Total rows: 3 Query complete 00:00:00.159 CRLF Ln 288, Col 22

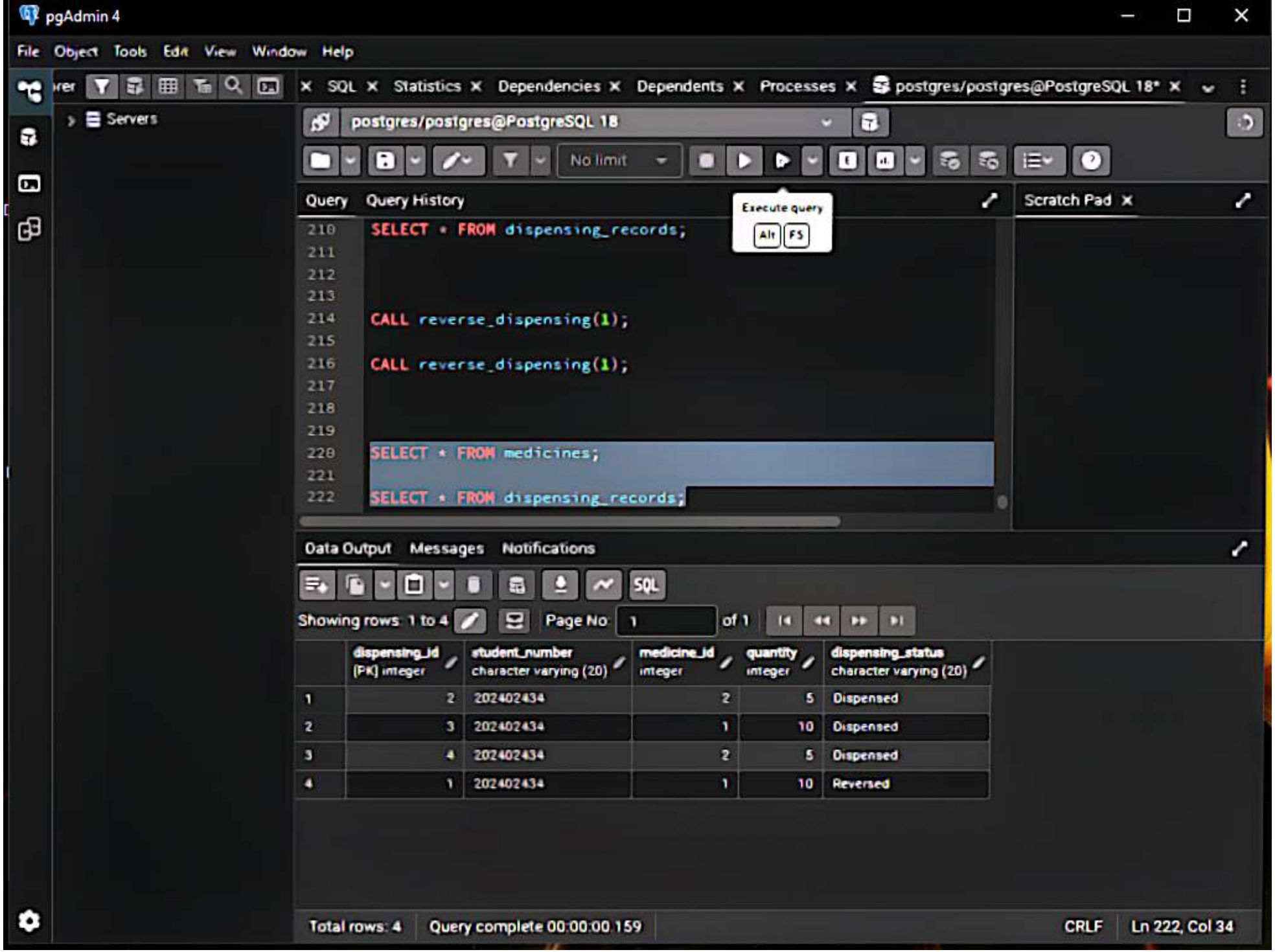
CamScanner


pgAdmin 4
File Object Tools Edit View Window Help
SQL Statistics Dependencies Dependents Processes postgres/postgres@PostgreSQL 18*
Servers
postgres/postgres@PostgreSQL 18
No limit
Query Query History
Execute query
Alt F5
Scratch Pad
264
265 IF v_quantity < 0 THEN
266 RAISE EXCEPTION
267 'Invalid quantity: dispensing quantity cannot be nega
268 END IF;
269
270 CALL dispense_medicine('202402434', 1, v_quantity);
271
272 EXCEPTION
273 WHEN OTHERS THEN
274 RAISE NOTICE 'Error handled: %', SQLERRM;
275 END;
276 $$;
Data Output Messages Notifications
NOTICE: Error handled: Invalid quantity: dispensing quantity cannot be negative
DO
Query returned successfully in 303 msec.
Total rows:
Query complete 00:00:00.303
CRLF
Ln 276, Col 4

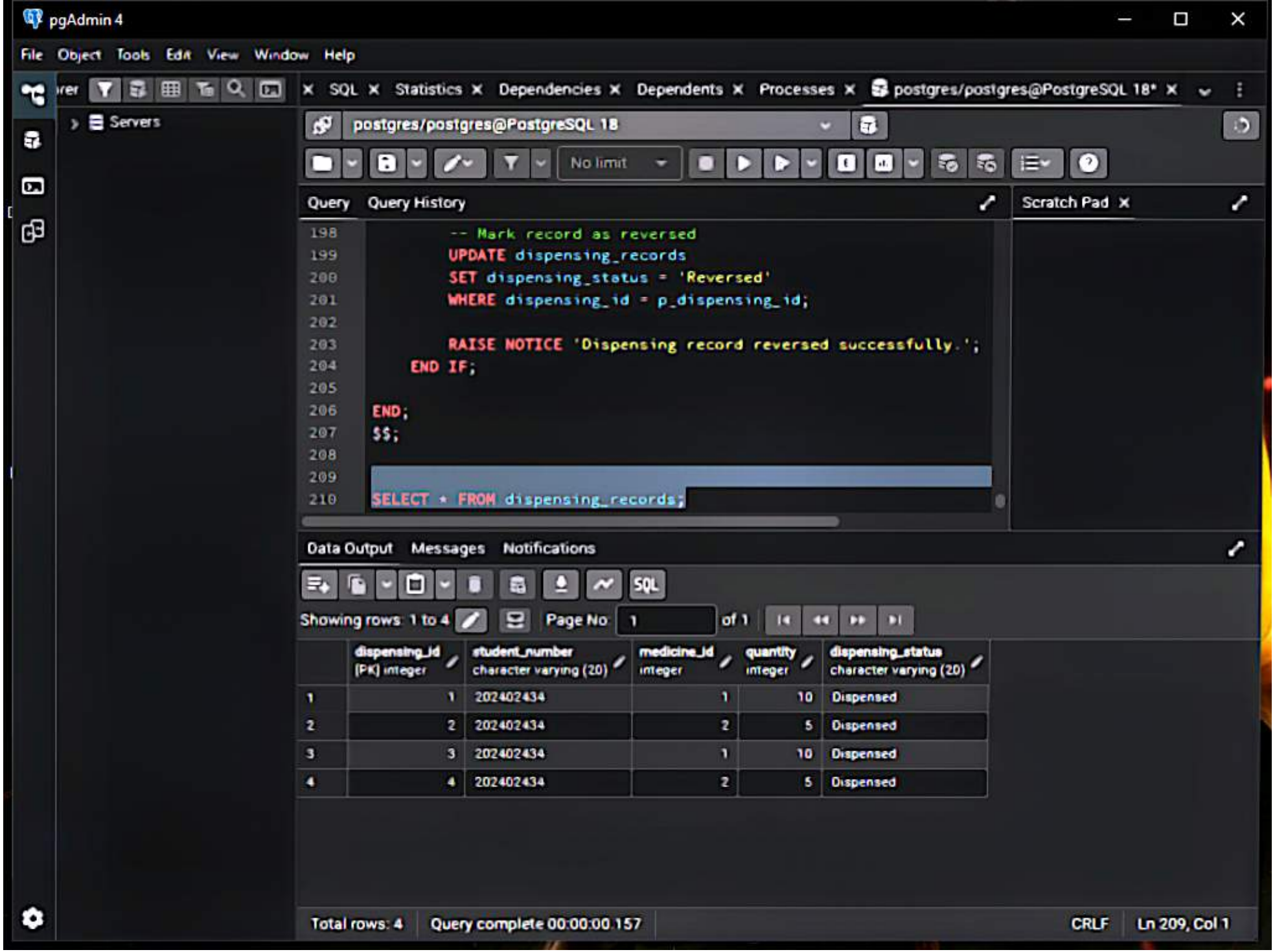
CamScanner


Help
SQL
Statistics
Dependencies
Dependents
Processes
postgres/postgres@PostgreSQL 18*
postgres/postgres@PostgreSQL 18
No limit
Query
Query History
Execute query
Alt F5
Scratch Pad
243
244 EXIT WHEN NOT FOUND;
245
246 RAISE NOTICE
247 'Medicine ID: %, Medicine: %, Stock Remaining: %',
248 medicine_record.medicine_id,
249 medicine_record.medicine_name,
250 medicine_record.stock_quantity;
251 END LOOP;
252
253 CLOSE medicine_cursor;
254 END;
255 $$;
Data Output
Messages
Notifications
oo
Query returned successfully in 159 msec.
Total rows:
Query complete 00:00:00.159
CRLF
Ln 255, Col 4

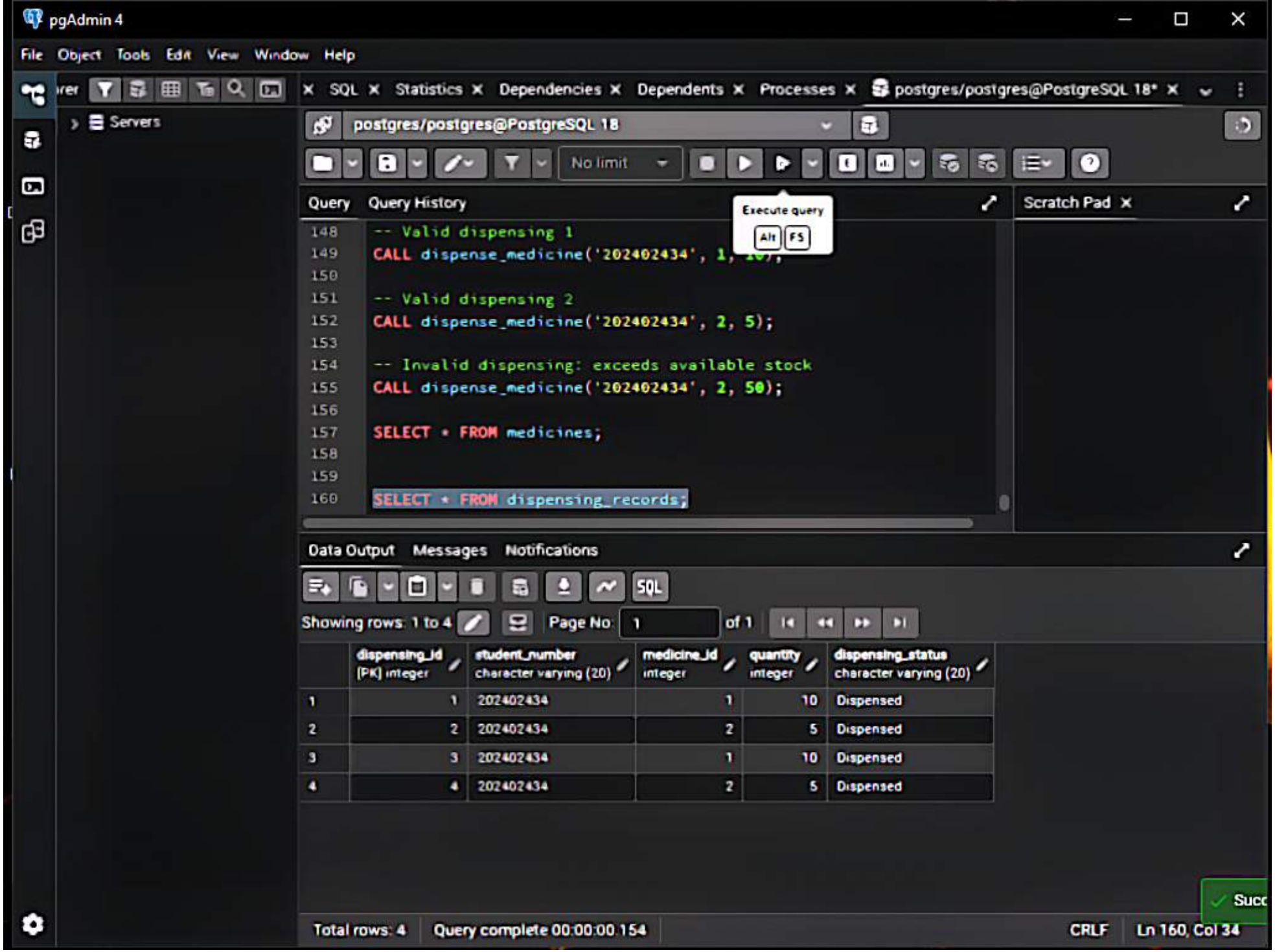
CamScanner


pgAdmin 4
File Object Tools Edit View Window Help
SQL Statistics Dependencies Dependents Processes postgres/postgres@PostgreSQL 18*
Servers
postgres/postgres@PostgreSQL 18
No limit
Query Query History
Execute query
Alt F5
Scratch Pad
210 SELECT * FROM dispensing_records;
211
212
213
214 CALL reverse_dispensing(1);
215
216 CALL reverse_dispensing(1);
217
218
219
220 SELECT * FROM medicines;
221
222 SELECT * FROM dispensing_records;
Data Output Messages Notifications
SQL
Showing rows: 1 to 4 Page No: 1 of 1
dispensing_id (PK) integer
student_number character varying (20)
medicine_id integer
quantity integer
dispensing_status character varying (20)
1 2 202402434 2 5 Dispensed
2 3 202402434 1 10 Dispensed
3 4 202402434 2 5 Dispensed
4 1 202402434 1 10 Reversed
Total rows: 4
Query complete 00:00:00.159
CRLF
Ln 222, Col 34

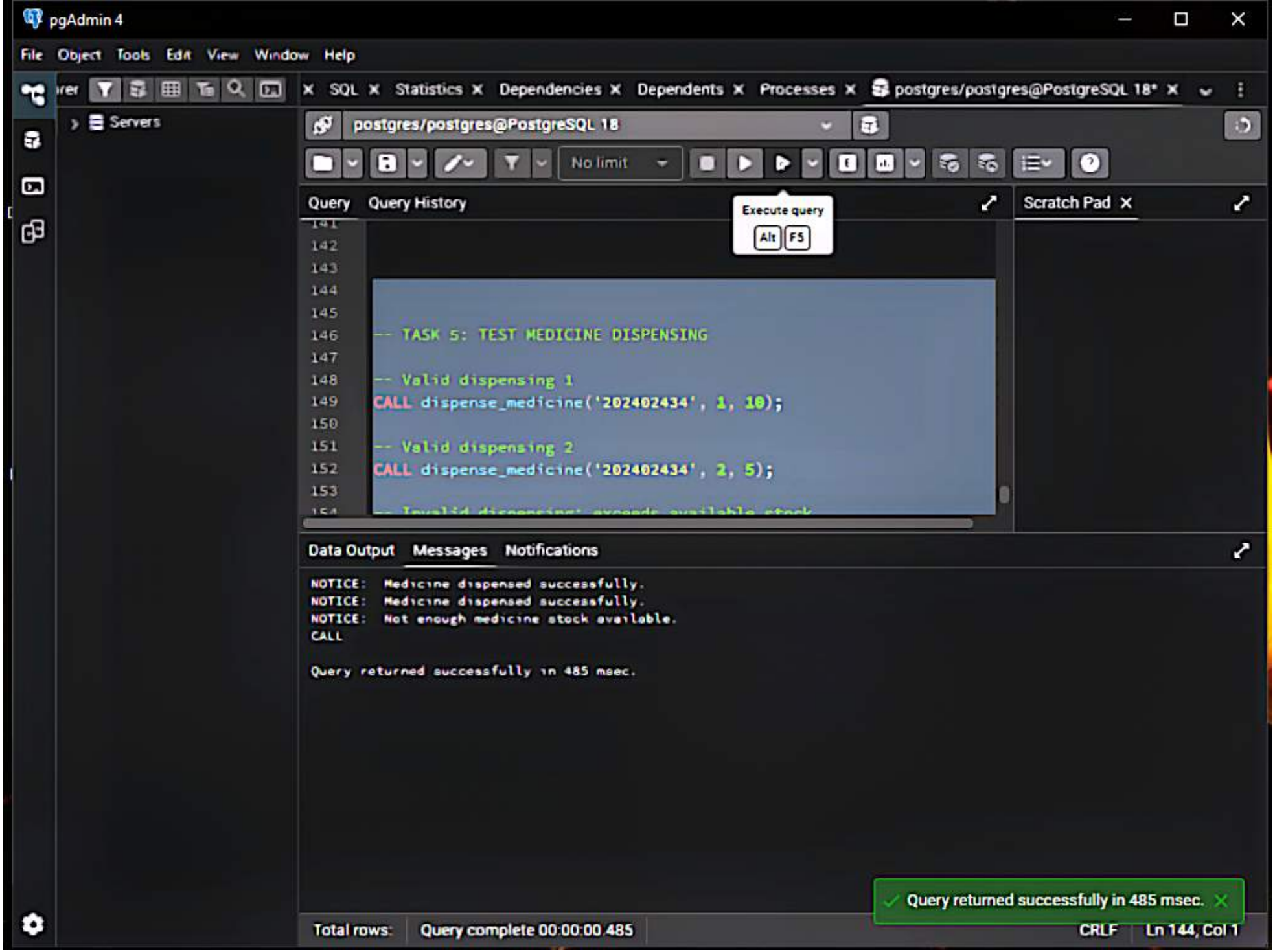
CamScanner


pgAdmin 4
File Object Tools Edit View Window Help
SQL Statistics Dependencies Dependents Processes postgres/postgres@PostgreSQL 18*
Servers
postgres/postgres@PostgreSQL 18
No limit
Query Query History
Scratch Pad
198 -- Mark record as reversed
199 UPDATE dispensing_records
200 SET dispensing_status = 'Reversed'
201 WHERE dispensing_id = p_dispensing_id;
202
203 RAISE NOTICE 'Dispensing record reversed successfully.';
204 END IF;
205
206 END;
207 $$;
208
209
210 SELECT * FROM dispensing_records;
Data Output Messages Notifications
SQL
Showing rows: 1 to 4 Page No 1 of 1
dispensing_id (PK) integer
student_number character varying (20)
medicine_id integer
quantity integer
dispensing_status character varying (20)
1 1 202402434 1 10 Dispensed
2 2 202402434 2 5 Dispensed
3 3 202402434 1 10 Dispensed
4 4 202402434 2 5 Dispensed
Total rows: 4
Query complete 00:00:00.157
CRLF
Ln 209, Col 1

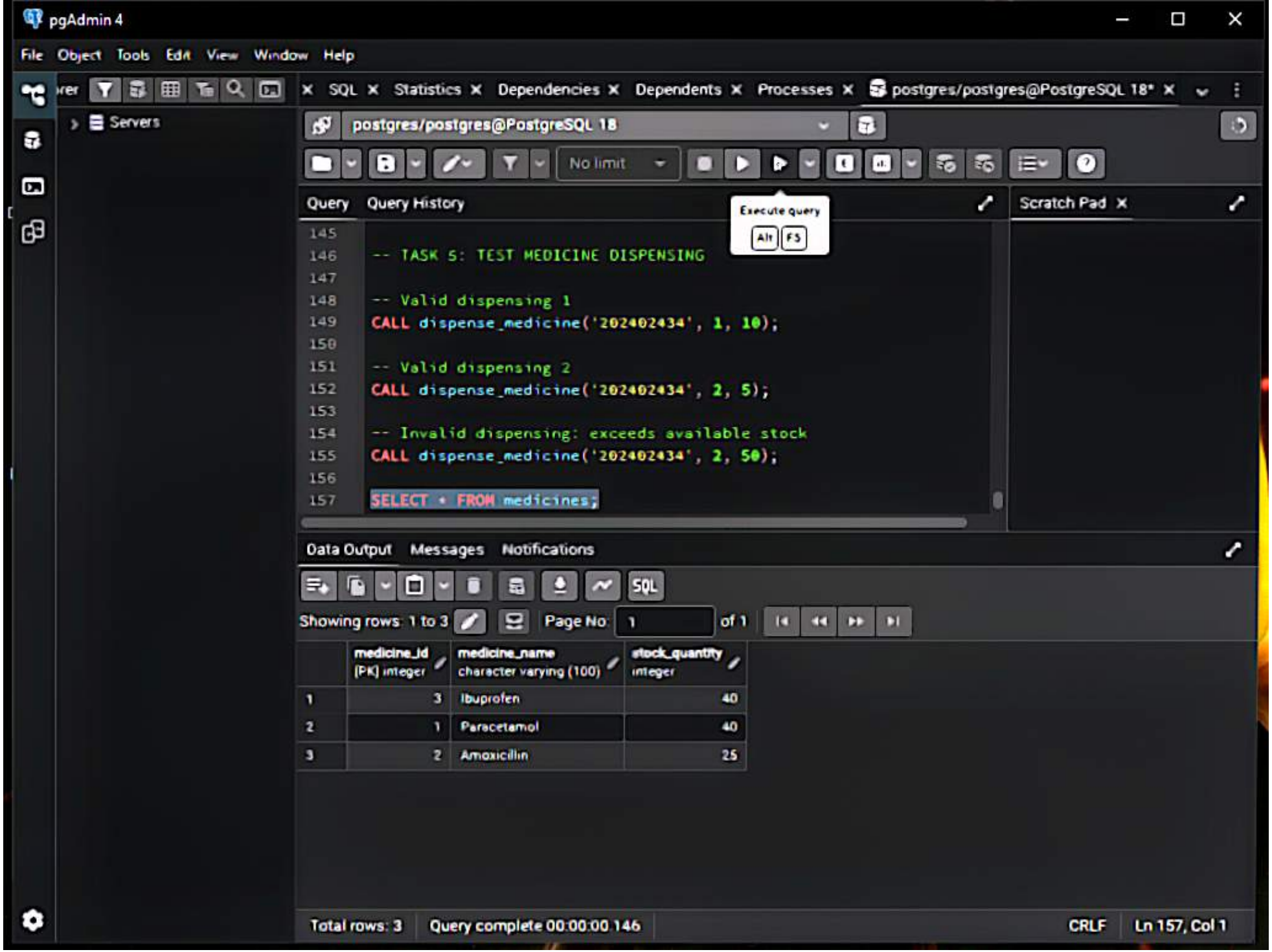
CamScanner


pgAdmin 4
File Object Tools Edit View Window Help
SQL Statistics Dependencies Dependents Processes postgres/postgres@PostgreSQL 18*
Servers
postgres/postgres@PostgreSQL 18
No limit
Query Query History
Execute query
Alt F5
Scratch Pad
148 -- Valid dispensing 1
149 CALL dispense_medicine('202402434', 1, 10);
150
151 -- Valid dispensing 2
152 CALL dispense_medicine('202402434', 2, 5);
153
154 -- Invalid dispensing: exceeds available stock
155 CALL dispense_medicine('202402434', 2, 50);
156
157 SELECT * FROM medicines;
158
159
160 SELECT * FROM dispensing_records;
Data Output Messages Notifications
SQL
Showing rows: 1 to 4 Page No: 1 of 1
dispensing_id (PK) integer
student_number character varying (20)
medicine_id integer
quantity integer
dispensing_status character varying (20)
1 1 202402434 1 10 Dispensed
2 2 202402434 2 5 Dispensed
3 3 202402434 1 10 Dispensed
4 4 202402434 2 5 Dispensed
Total rows: 4
Query complete 00:00:00.154
CRLF
Ln 160, Col 34

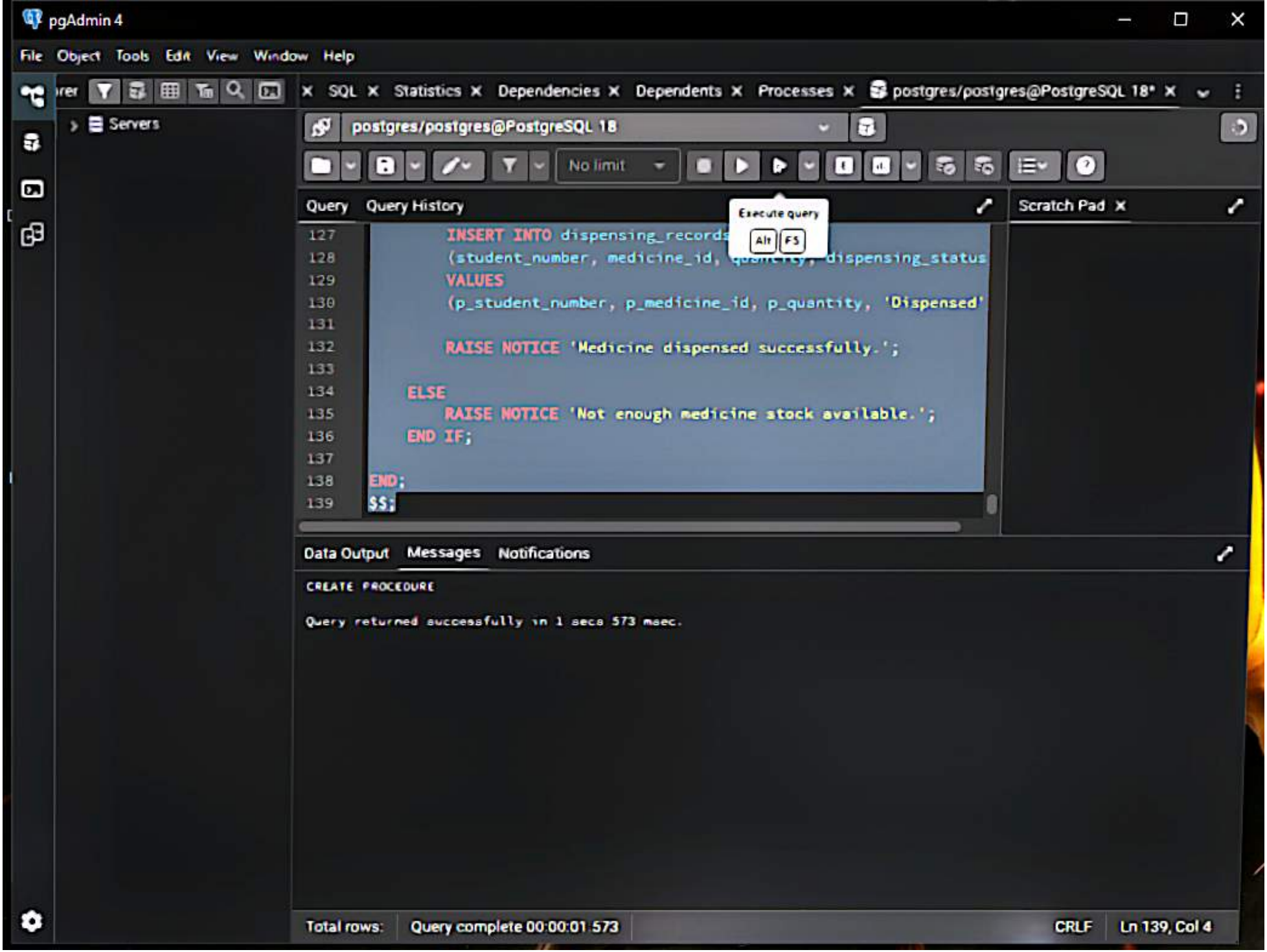
CamScanner


pgAdmin 4
File Object Tools Edit View Window Help
SQL Statistics Dependencies Dependents Processes postgres/postgres@PostgreSQL 18*
Servers
postgres/postgres@PostgreSQL 18
No limit
Query Query History
Execute query
Alt F5
Scratch Pad
141
142
143
144
145
146 -- TASK 5: TEST MEDICINE DISPENSING
147
148 -- Valid dispensing 1
149 CALL dispense_medicine('202402434', 1, 10);
150
151 -- Valid dispensing 2
152 CALL dispense_medicine('202402434', 2, 5);
153
Data Output Messages Notifications
NOTICE: Medicine dispensed successfully.
NOTICE: Medicine dispensed successfully.
NOTICE: Not enough medicine stock available.
CALL
Query returned successfully in 485 msec.
Query returned successfully in 485 msec.
Total rows:
Query complete 00:00:00.485
CRLF
Ln 144, Col 1

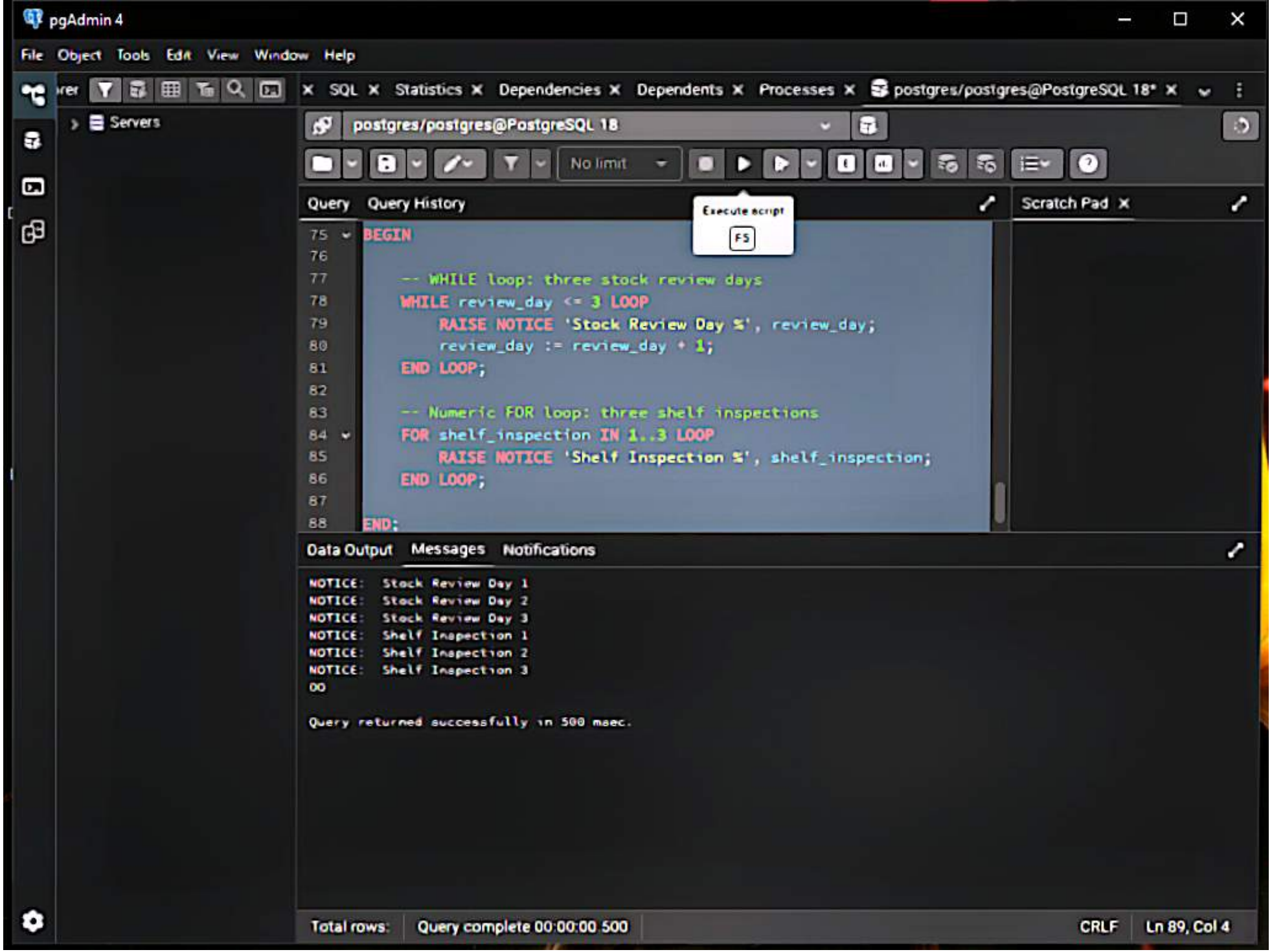
CamScanner


pgAdmin 4
File Object Tools Edit View Window Help
SQL Statistics Dependencies Dependents Processes postgres/postgres@PostgreSQL 18*
Servers
postgres/postgres@PostgreSQL 18
No limit
Query Query History
Execute query
Alt F5
Scratch Pad
145
146 -- TASK 5: TEST MEDICINE DISPENSING
147
148 -- Valid dispensing 1
149 CALL dispense_medicine('202402434', 1, 10);
150
151 -- Valid dispensing 2
152 CALL dispense_medicine('202402434', 2, 5);
153
154 -- Invalid dispensing: exceeds available stock
155 CALL dispense_medicine('202402434', 2, 50);
156
157 SELECT * FROM medicines;
Data Output Messages Notifications
SQL
Showing rows: 1 to 3 Page No: 1 of 1
medicine_id (PK) integer
medicine_name character varying (100)
stock_quantity integer
1 3 Ibuprofen 40
2 1 Paracetamol 40
3 2 Amoxicillin 25
Total rows: 3 Query complete 00:00:00.146 CRLF Ln 157, Col 1

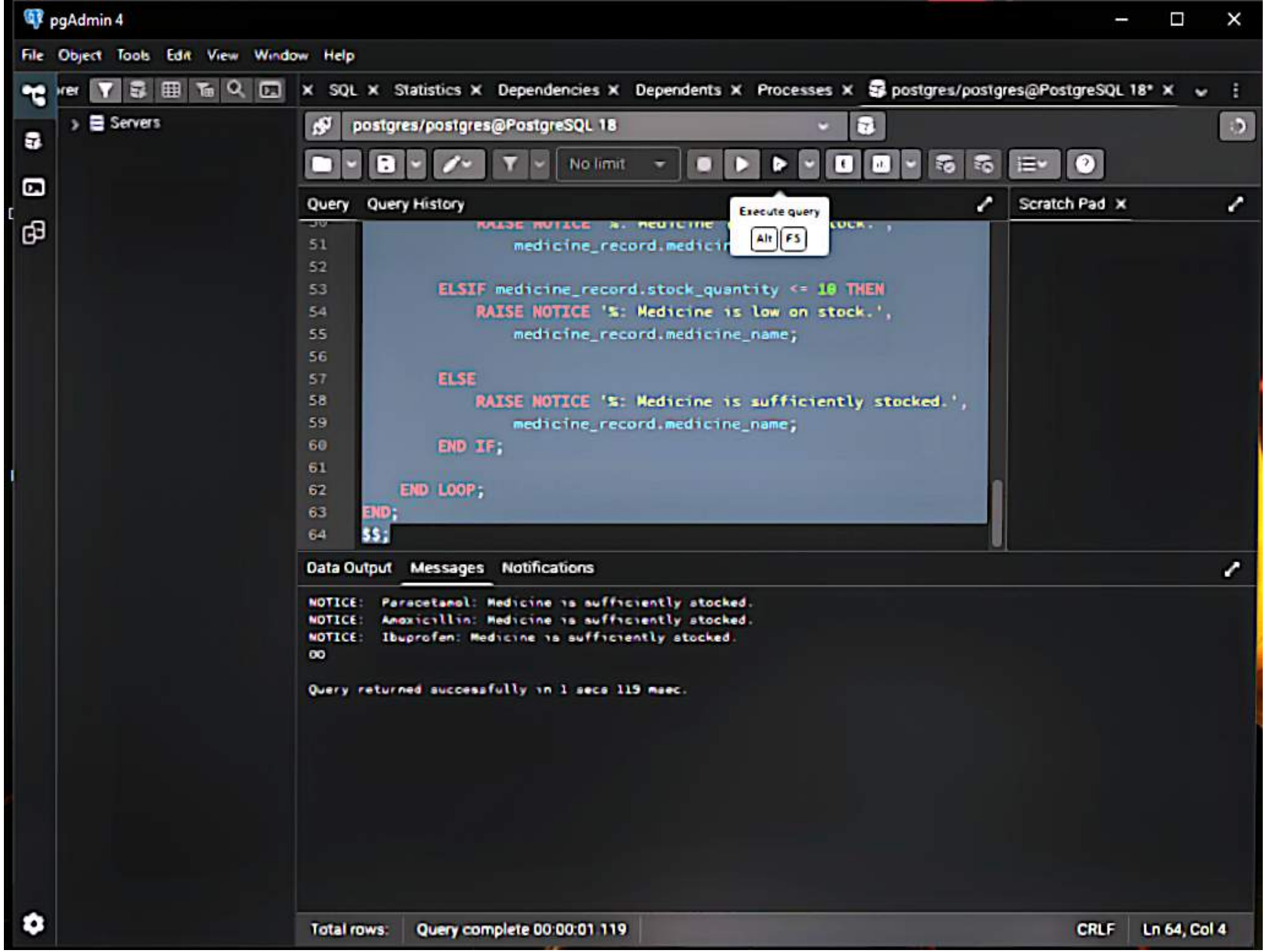
CamScanner


pgAdmin 4
File Object Tools Edit View Window Help
SQL Statistics Dependencies Dependents Processes postgres/postgres@PostgreSQL 18*
Servers
postgres/postgres@PostgreSQL 18
No limit
Query Query History
Execute query
Alt F5
Scratch Pad
127 INSERT INTO dispensing_records
128 (student_number, medicine_id, quantity, dispensing_status
129 VALUES
130 (p_student_number, p_medicine_id, p_quantity, 'Dispensed'
131
132 RAISE NOTICE 'Medicine dispensed successfully.';
133
134 ELSE
135 RAISE NOTICE 'Not enough medicine stock available.';
136 END IF;
137
138 END;
139 $$;
Data Output Messages Notifications
CREATE PROCEDURE
Query returned successfully in 1 secs 573 msec.
Total rows:
Query complete 00:00:01.573
CRLF
Ln 139, Col 4

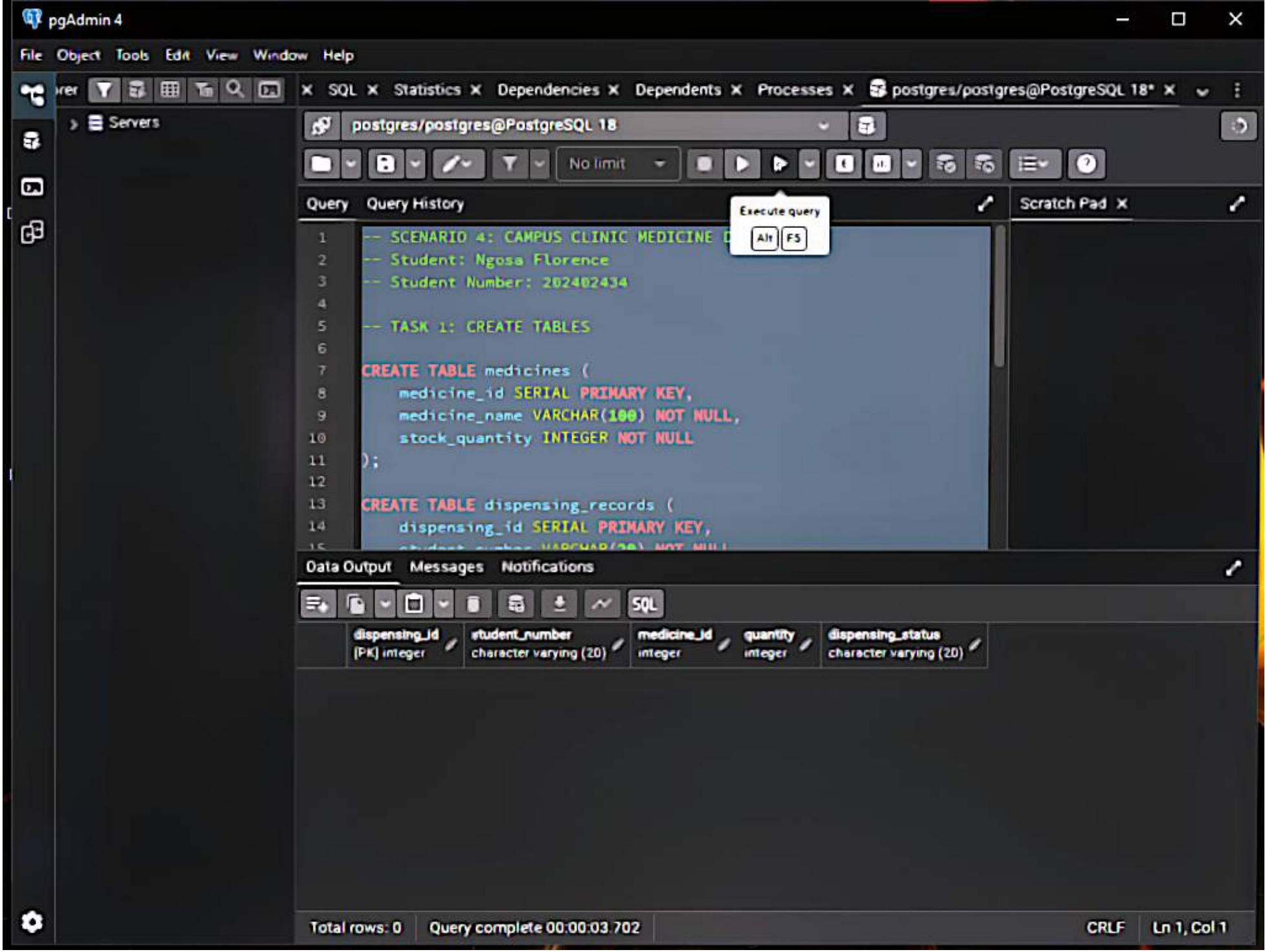
CamScanner


pgAdmin 4
File Object Tools Edit View Window Help
SQL Statistics Dependencies Dependents Processes postgres/postgres@PostgreSQL 18*
Servers
postgres/postgres@PostgreSQL 18
No limit
Query Query History
Execute script
F5
Scratch Pad
75 BEGIN
76
77 -- WHILE loop: three stock review days
78 WHILE review_day <= 3 LOOP
79 RAISE NOTICE 'Stock Review Day %', review_day;
80 review_day := review_day + 1;
81 END LOOP;
82
83 -- Numeric FOR loop: three shelf inspections
84 FOR shelf_inspection IN 1..3 LOOP
85 RAISE NOTICE 'Shelf Inspection %', shelf_inspection;
86 END LOOP;
87
88 END;
Data Output Messages Notifications
NOTICE: Stock Review Day 1
NOTICE: Stock Review Day 2
NOTICE: Stock Review Day 3
NOTICE: Shelf Inspection 1
NOTICE: Shelf Inspection 2
NOTICE: Shelf Inspection 3
DO
Query returned successfully in 500 msec.
Total rows:
Query complete 00:00:00.500
CRLF
Ln 89, Col 4

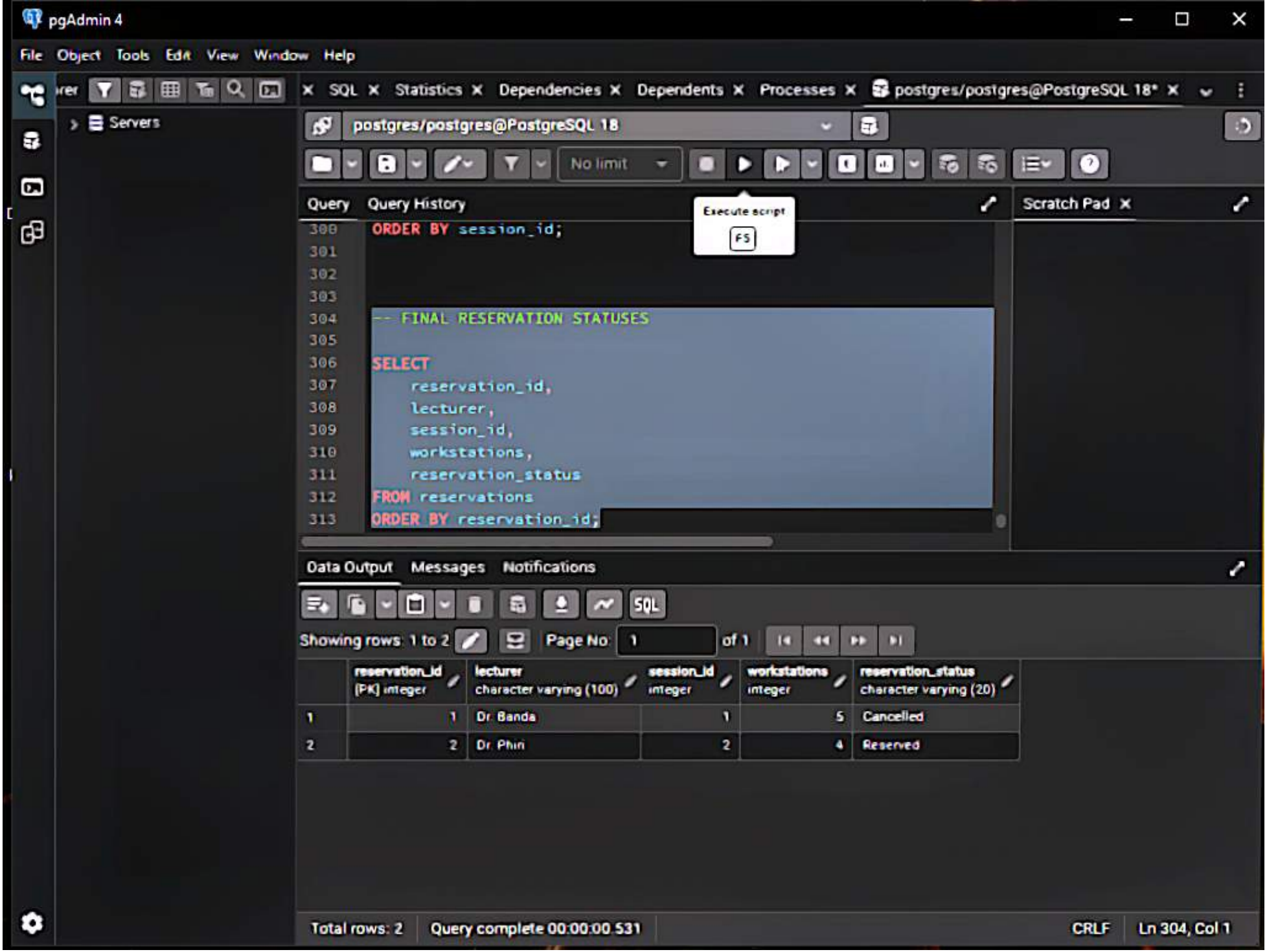
CamScanner


pgAdmin 4
File Object Tools Edit View Window Help
SQL Statistics Dependencies Dependents Processes postgres/postgres@PostgreSQL 18*
Servers
postgres/postgres@PostgreSQL 18
No limit
Query Query History
Execute query
Alt F5
Scratch Pad
51 medicine_record.medicine_name;
52
53 ELSIF medicine_record.stock_quantity <= 10 THEN
54 RAISE NOTICE '%: Medicine is low on stock.',
55 medicine_record.medicine_name;
56
57 ELSE
58 RAISE NOTICE '%: Medicine is sufficiently stocked.',
59 medicine_record.medicine_name;
60 END IF;
61
62 END LOOP;
63 END;
64 $$;
Data Output Messages Notifications
NOTICE: Paracetamol: Medicine is sufficiently stocked.
NOTICE: Amoxicillin: Medicine is sufficiently stocked.
NOTICE: Ibuprofen: Medicine is sufficiently stocked.
DO
Query returned successfully in 1 secs 119 msec.
Total rows:
Query complete 00:00:01.119
CRLF
Ln 64, Col 4

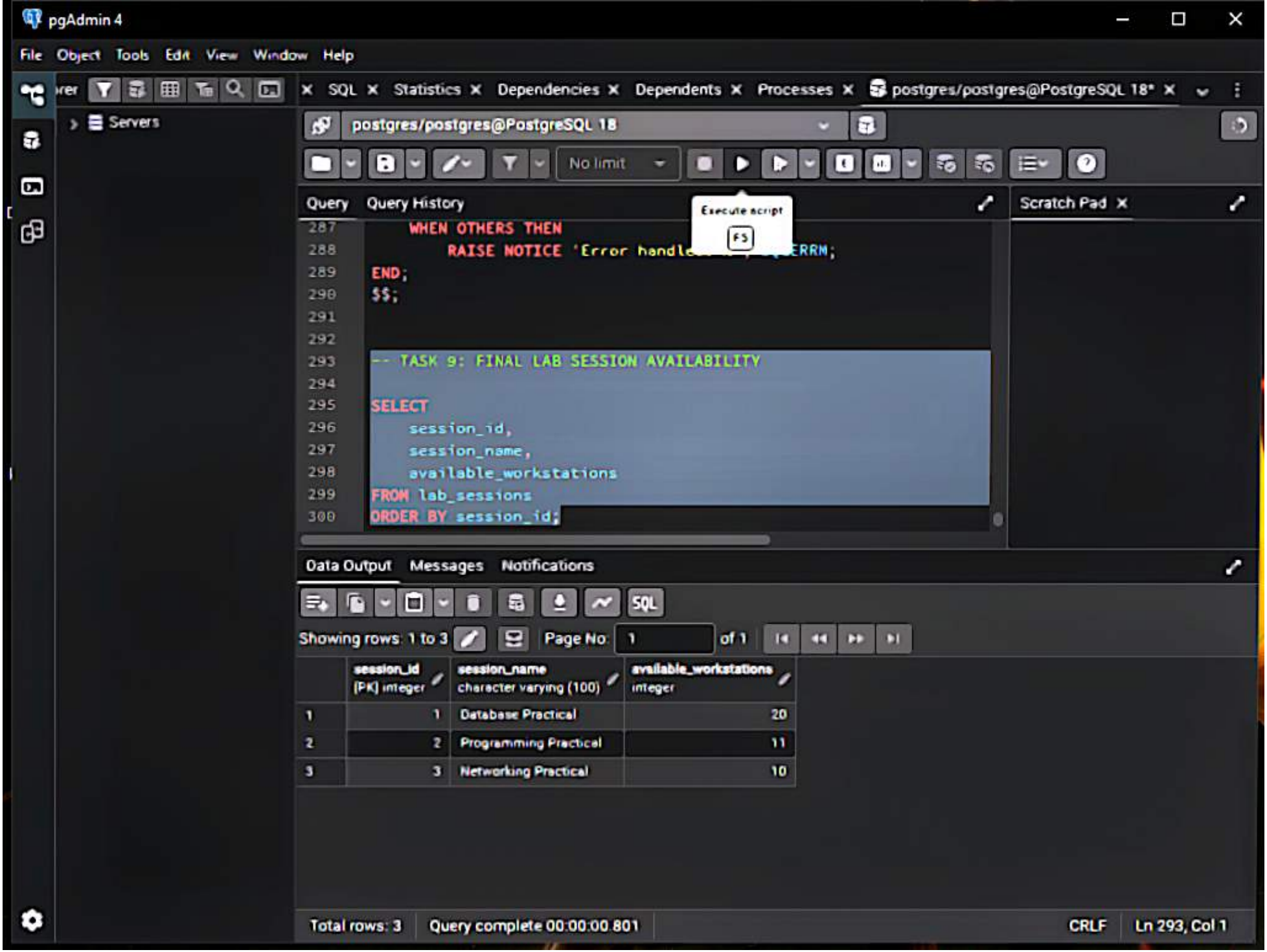
CamScanner


pgAdmin 4
File Object Tools Edit View Window Help
Servers
SQL Statistics Dependencies Dependents Processes postgres/postgres@PostgreSQL 18*
postgres/postgres@PostgreSQL 18
No limit
Query Query History
Execute query
Alt F5
Scratch Pad
1 -- SCENARIO 4: CAMPUS CLINIC MEDICINE D
2 -- Student: Ngosa Florence
3 -- Student Number: 202402434
4
5 -- TASK 1: CREATE TABLES
6
7 CREATE TABLE medicines (
8 medicine_id SERIAL PRIMARY KEY,
9 medicine_name VARCHAR(100) NOT NULL,
10 stock_quantity INTEGER NOT NULL
11);
12
13 CREATE TABLE dispensing_records (
14 dispensing_id SERIAL PRIMARY KEY,
Data Output Messages Notifications
SQL
dispensing_id (PK) integer
student_number character varying (20)
medicine_id integer
quantity integer
dispensing_status character varying (20)
Total rows: 0
Query complete 00:00:03.702
CRLF
Ln 1, Col 1

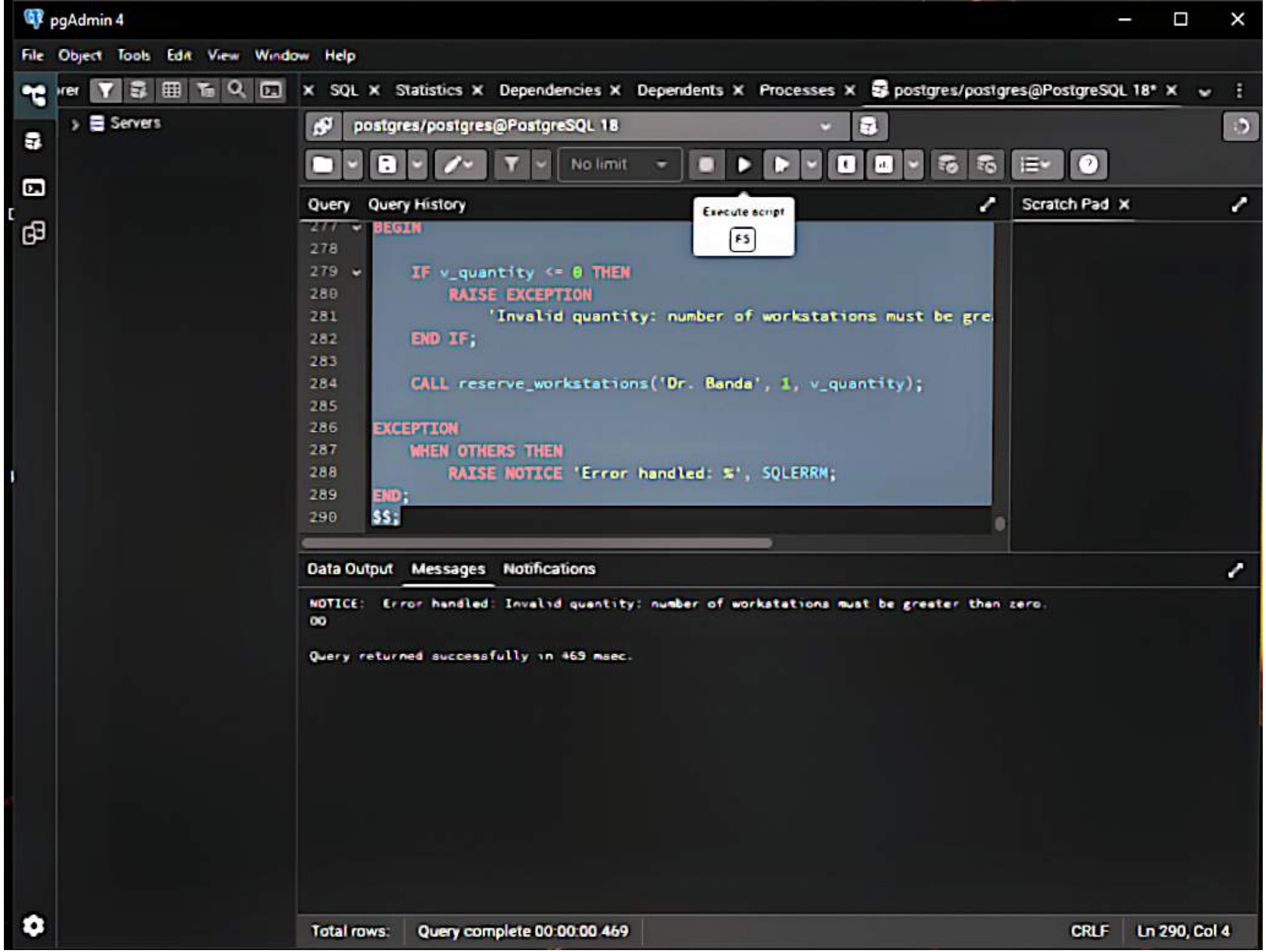
CamScanner


pgAdmin 4
File Object Tools Edit View Window Help
SQL Statistics Dependencies Dependents Processes postgres/postgres@PostgreSQL 18*
Servers
postgres/postgres@PostgreSQL 18
No limit
Query Query History
Execute script
F5
Scratch Pad
300 ORDER BY session_id;
301
302
303
304 -- FINAL RESERVATION STATUSES
305
306 SELECT
307 reservation_id,
308 lecturer,
309 session_id,
310 workstations,
311 reservation_status
312 FROM reservations
313 ORDER BY reservation_id;
Data Output Messages Notifications
SQL
Showing rows: 1 to 2 Page No: 1 of 1
reservation_id (PK) integer
lecturer character varying (100)
session_id integer
workstations integer
reservation_status character varying (20)
1 1 Dr. Banda 1 5 Cancelled
2 2 Dr. Phiri 2 4 Reserved
Total rows: 2 Query complete 00:00:00.531 CRLF Ln 304, Col 1

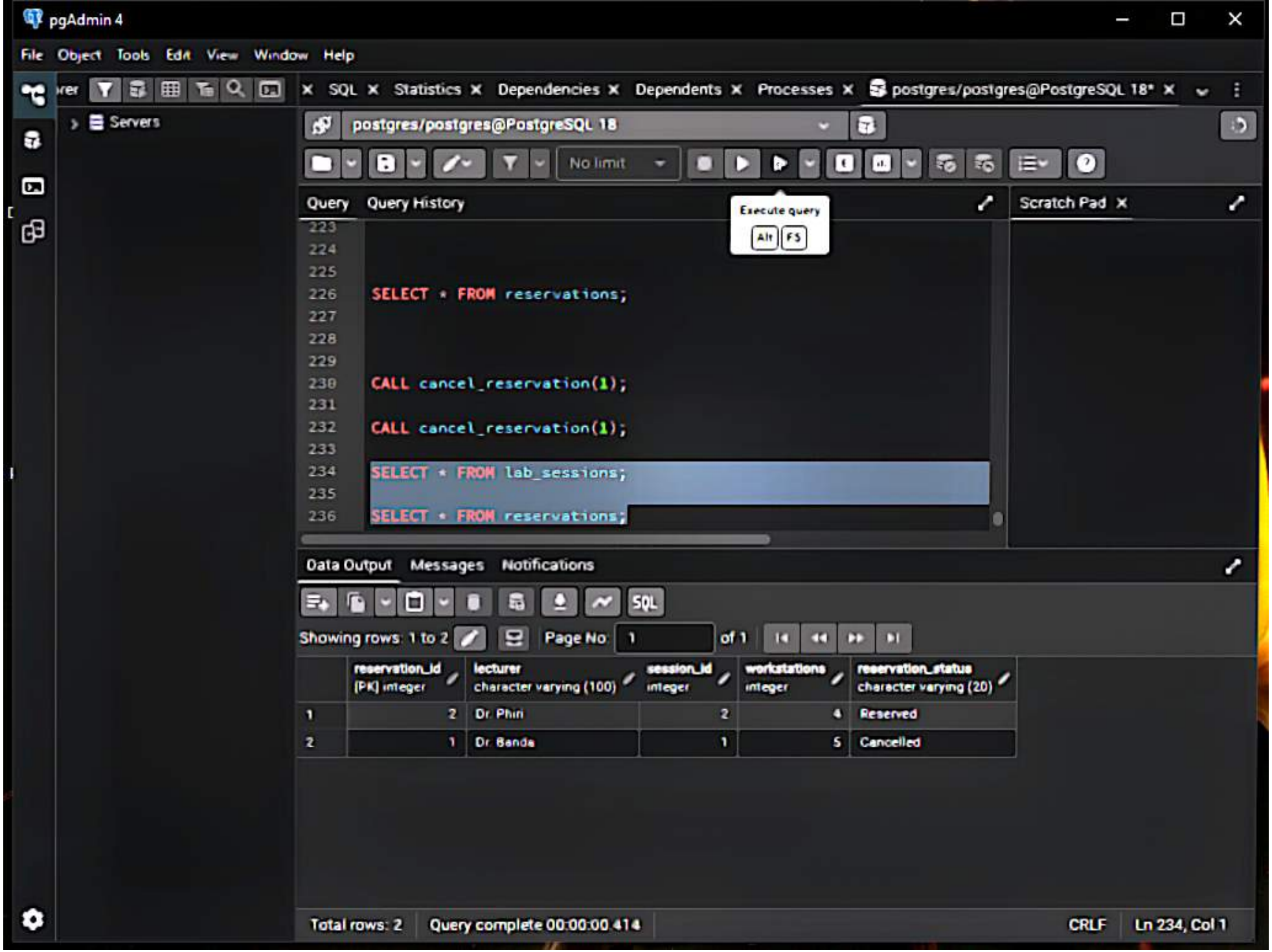
CamScanner


pgAdmin 4
File Object Tools Edit View Window Help
Servers
SQL Statistics Dependencies Dependents Processes postgres/postgres@PostgreSQL 18*
postgres/postgres@PostgreSQL 18
No limit
Query Query History
Execute script
F5
Scratch Pad
287 WHEN OTHERS THEN
288 RAISE NOTICE 'Error handle ERRM;
289 END;
290 $$;
291
292
293 -- TASK 9: FINAL LAB SESSION AVAILABILITY
294
295 SELECT
296 session_id,
297 session_name,
298 available_workstations
299 FROM lab_sessions
300 ORDER BY session_id;
Data Output Messages Notifications
SQL
Showing rows: 1 to 3 Page No: 1 of 1
session_id (PK) integer
session_name character varying (100)
available_workstations integer
1 1 Database Practical 20
2 2 Programming Practical 11
3 3 Networking Practical 10
Total rows: 3 Query complete 00:00:00.801 CRLF Ln 293, Col 1

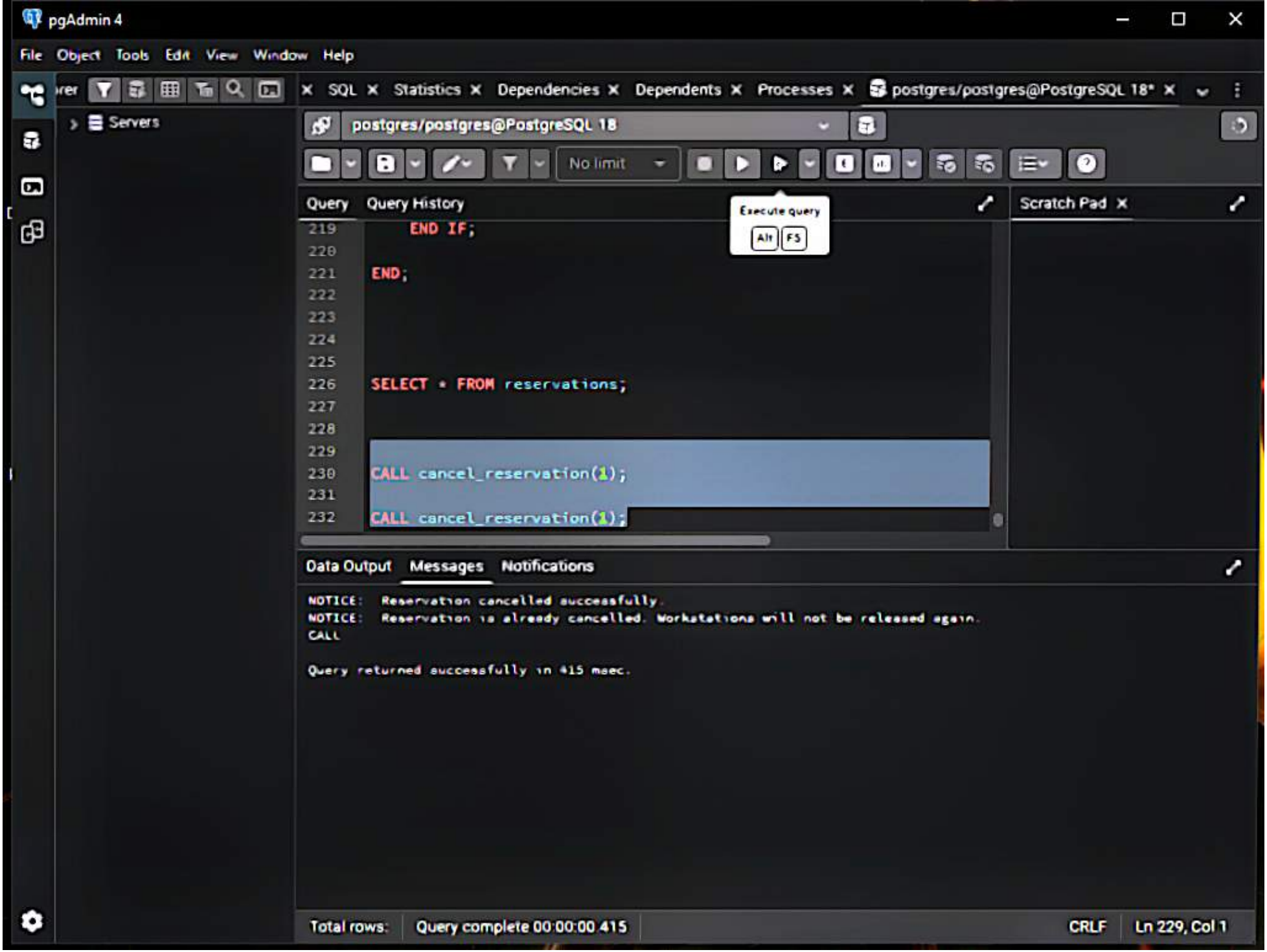
CamScanner


pgAdmin 4
File Object Tools Edit View Window Help
SQL Statistics Dependencies Dependents Processes postgres/postgres@PostgreSQL 18*
Servers
postgres/postgres@PostgreSQL 18
No limit
Query Query History
Execute script
F5
Scratch Pad
277 BEGIN
278
279 IF v_quantity <= 0 THEN
280 RAISE EXCEPTION
281 'Invalid quantity: number of workstations must be gre
282 END IF;
283
284 CALL reserve_workstations('Dr. Banda', 1, v_quantity);
285
286 EXCEPTION
287 WHEN OTHERS THEN
288 RAISE NOTICE 'Error handled: %', SQLERRM;
289 END;
290 $$;
Data Output Messages Notifications
NOTICE: Error handled: Invalid quantity: number of workstations must be greater than zero.
DO
Query returned successfully in 469 msec.
Total rows:
Query complete 00:00:00.469
CRLF
Ln 290, Col 4

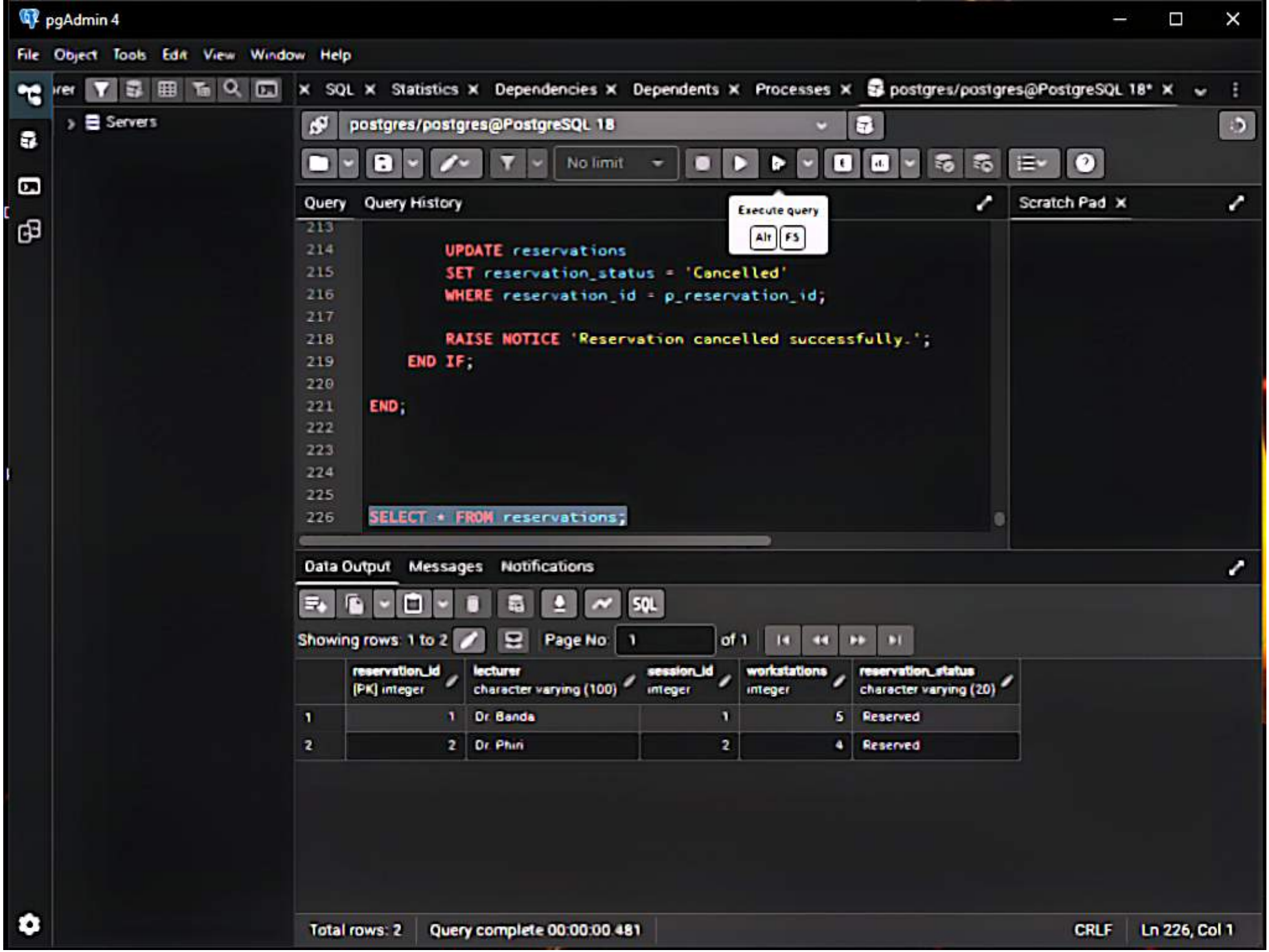
CamScanner


pgAdmin 4
File Object Tools Edit View Window Help
SQL Statistics Dependencies Dependents Processes postgres/postgres@PostgreSQL 18*
Servers
postgres/postgres@PostgreSQL 18
No limit
Query Query History
Execute query
Alt F5
Scratch Pad
223
224
225
226 SELECT * FROM reservations;
227
228
229
230 CALL cancel_reservation(1);
231
232 CALL cancel_reservation(1);
233
234 SELECT * FROM lab_sessions;
235
236 SELECT * FROM reservations;
Data Output Messages Notifications
SQL
Showing rows: 1 to 2 Page No: 1 of 1
reservation_id (PK) integer
lecturer character varying (100)
session_id integer
workstations integer
reservation_status character varying (20)
1 2 Dr. Phiri 2 4 Reserved
2 1 Dr. Banda 1 5 Cancelled
Total rows: 2 Query complete 00:00:00.414 CRLF Ln 234, Col 1

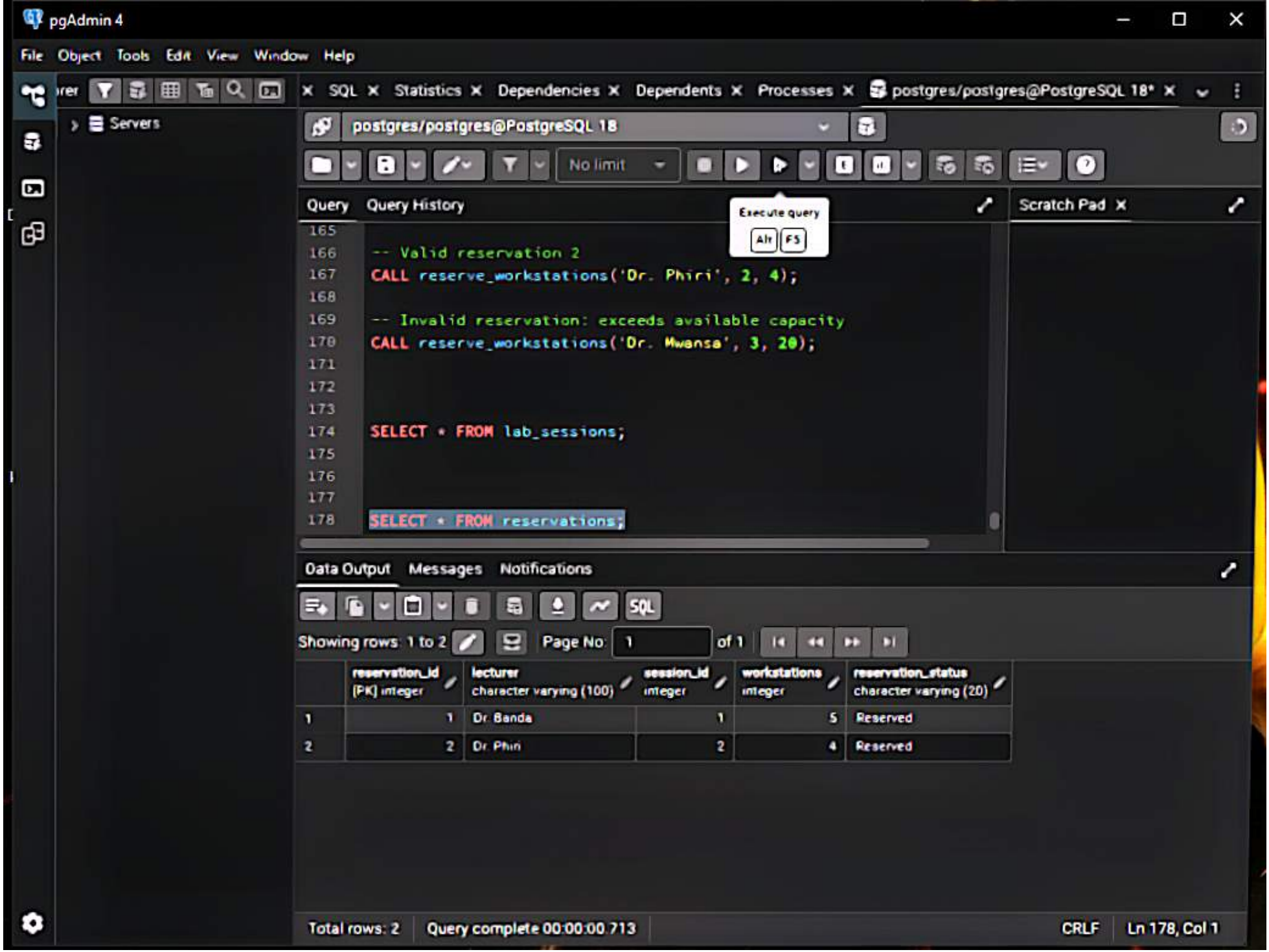
CamScanner


pgAdmin 4
File Object Tools Edit View Window Help
Servers
SQL Statistics Dependencies Dependents Processes postgres/postgres@PostgreSQL 18*
postgres/postgres@PostgreSQL 18
No limit
Query Query History
Execute query
Alt F5
Scratch Pad
219 END IF;
220
221 END;
222
223
224
225
226 SELECT * FROM reservations;
227
228
229
230 CALL cancel_reservation(1);
231
232 CALL cancel_reservation(1);
Data Output Messages Notifications
NOTICE: Reservation cancelled successfully.
NOTICE: Reservation is already cancelled. Workstations will not be released again.
CALL
Query returned successfully in 415 msec.
Total rows:
Query complete 00:00:00.415
CRLF
Ln 229, Col 1

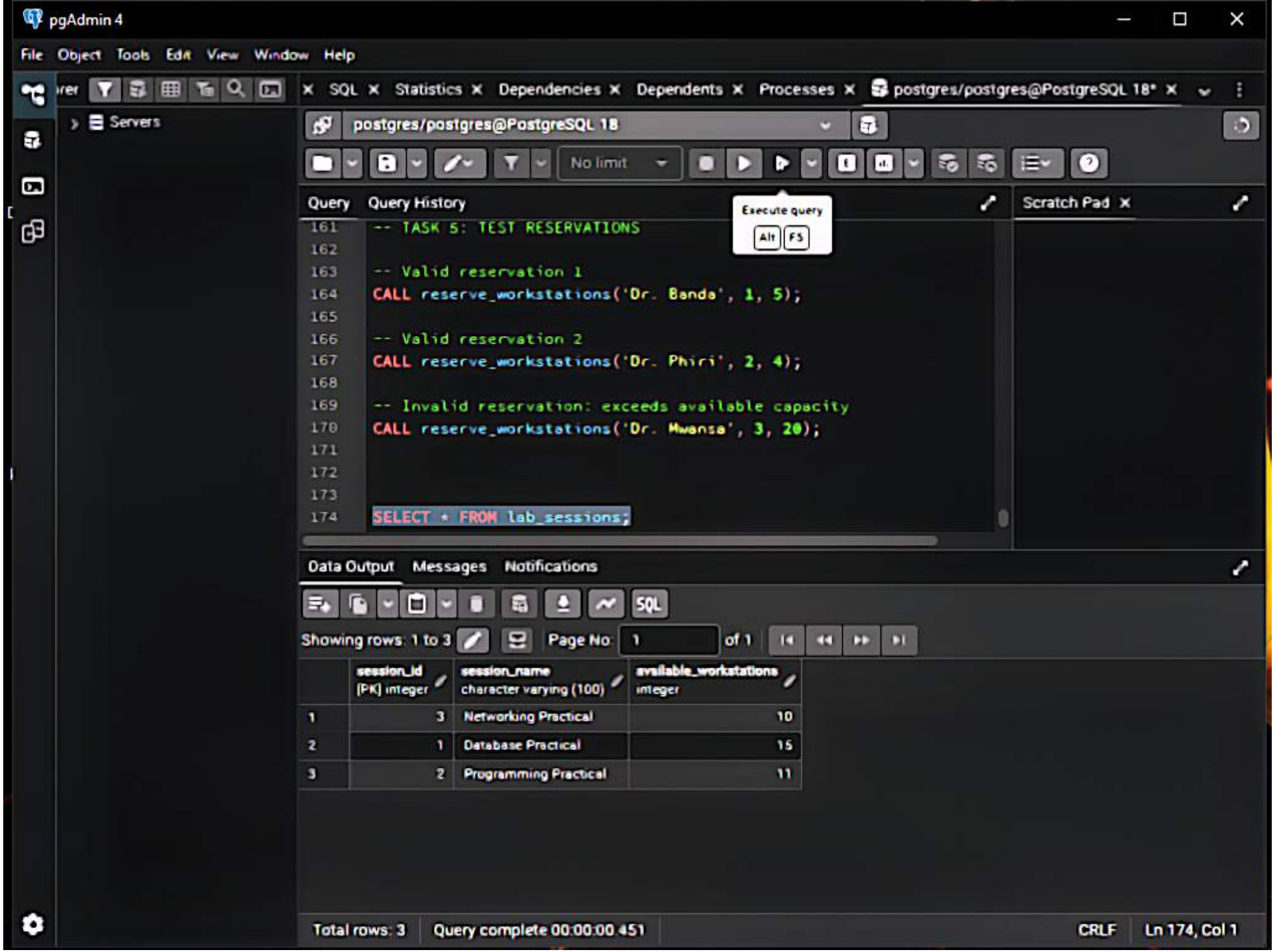
CamScanner


pgAdmin 4
File Object Tools Edit View Window Help
SQL Statistics Dependencies Dependents Processes postgres/postgres@PostgreSQL 18*
Servers
postgres/postgres@PostgreSQL 18
No limit
Query Query History
Execute query
Alt F5
Scratch Pad
213
214 UPDATE reservations
215 SET reservation_status = 'Cancelled'
216 WHERE reservation_id = p_reservation_id;
217
218 RAISE NOTICE 'Reservation cancelled successfully.';
219 END IF;
220
221 END;
222
223
224
225
226 SELECT * FROM reservations;
Data Output Messages Notifications
SQL
Showing rows: 1 to 2 Page No: 1 of 1
reservation_id [PK] integer
lecturer character varying (100)
session_id integer
workstations integer
reservation_status character varying (20)
1 1 Dr. Banda 1 5 Reserved
2 2 Dr. Phiri 2 4 Reserved
Total rows: 2 Query complete 00:00:00.481 CRLF Ln 226, Col 1

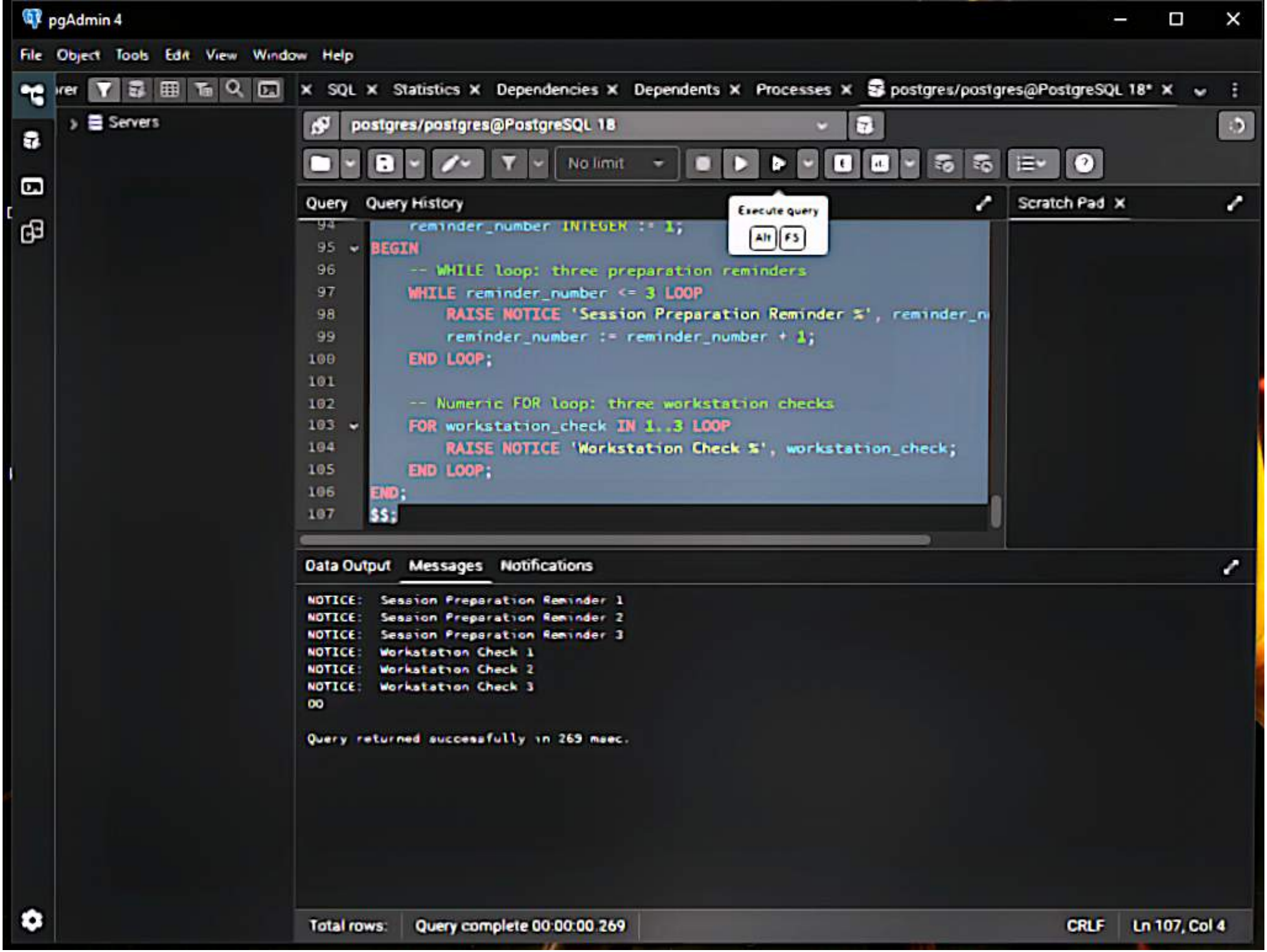
CamScanner


pgAdmin 4
File Object Tools Edit View Window Help
Servers
SQL Statistics Dependencies Dependents Processes postgres/postgres@PostgreSQL 18*
postgres/postgres@PostgreSQL 18
No limit
Query Query History
Execute query
Alt F5
Scratch Pad
165
166 -- Valid reservation 2
167 CALL reserve_workstations('Dr. Phiri', 2, 4);
168
169 -- Invalid reservation: exceeds available capacity
170 CALL reserve_workstations('Dr. Mwansa', 3, 20);
171
172
173
174 SELECT * FROM lab_sessions;
175
176
177
178 SELECT * FROM reservations;
Data Output Messages Notifications
SQL
Showing rows: 1 to 2 Page No: 1 of 1
reservation_id (PK) integer
lecturer character varying (100)
session_id integer
workstations integer
reservation_status character varying (20)
1 1 Dr. Banda 1 5 Reserved
2 2 Dr. Phiri 2 4 Reserved
Total rows: 2 Query complete 00:00:00.713 CRLF Ln 178, Col 1

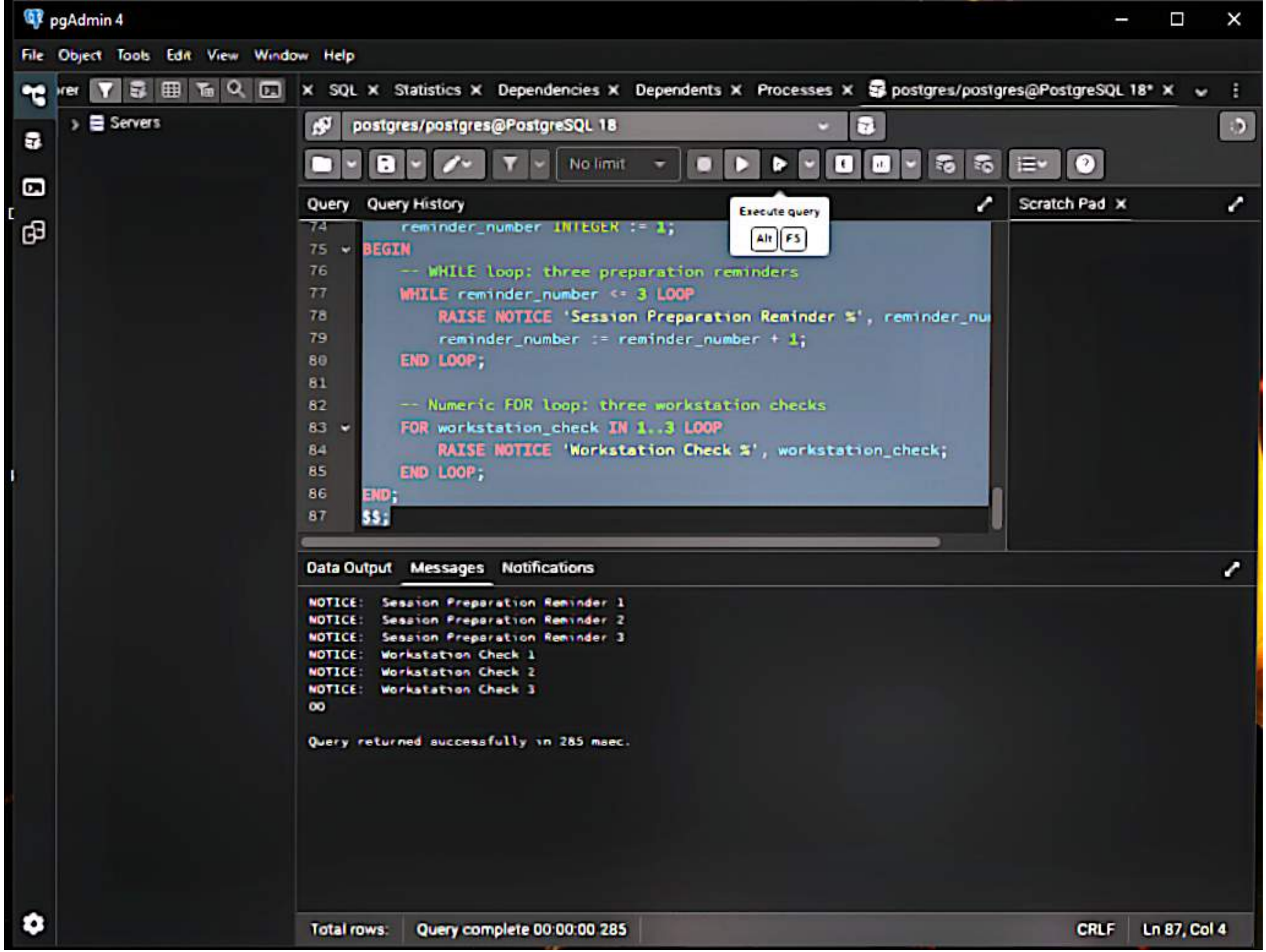
CamScanner


pgAdmin 4
File Object Tools Edit View Window Help
SQL Statistics Dependencies Dependents Processes postgres/postgres@PostgreSQL 18*
Servers
postgres/postgres@PostgreSQL 18
No limit
Query Query History
Execute query
Alt F5
Scratch Pad
161 -- TASK 5: TEST RESERVATIONS
162
163 -- Valid reservation 1
164 CALL reserve_workstations('Dr. Banda', 1, 5);
165
166 -- Valid reservation 2
167 CALL reserve_workstations('Dr. Phiri', 2, 4);
168
169 -- Invalid reservation: exceeds available capacity
170 CALL reserve_workstations('Dr. Mwansa', 3, 20);
171
172
173
174 SELECT * FROM lab_sessions;
Data Output Messages Notifications
SQL
Showing rows: 1 to 3 Page No: 1 of 1
session_id (PK) integer
session_name character varying (100)
available_workstations integer
1 3 Networking Practical 10
2 1 Database Practical 15
3 2 Programming Practical 11
Total rows: 3 Query complete 00:00:00.451 CRLF Ln 174, Col 1

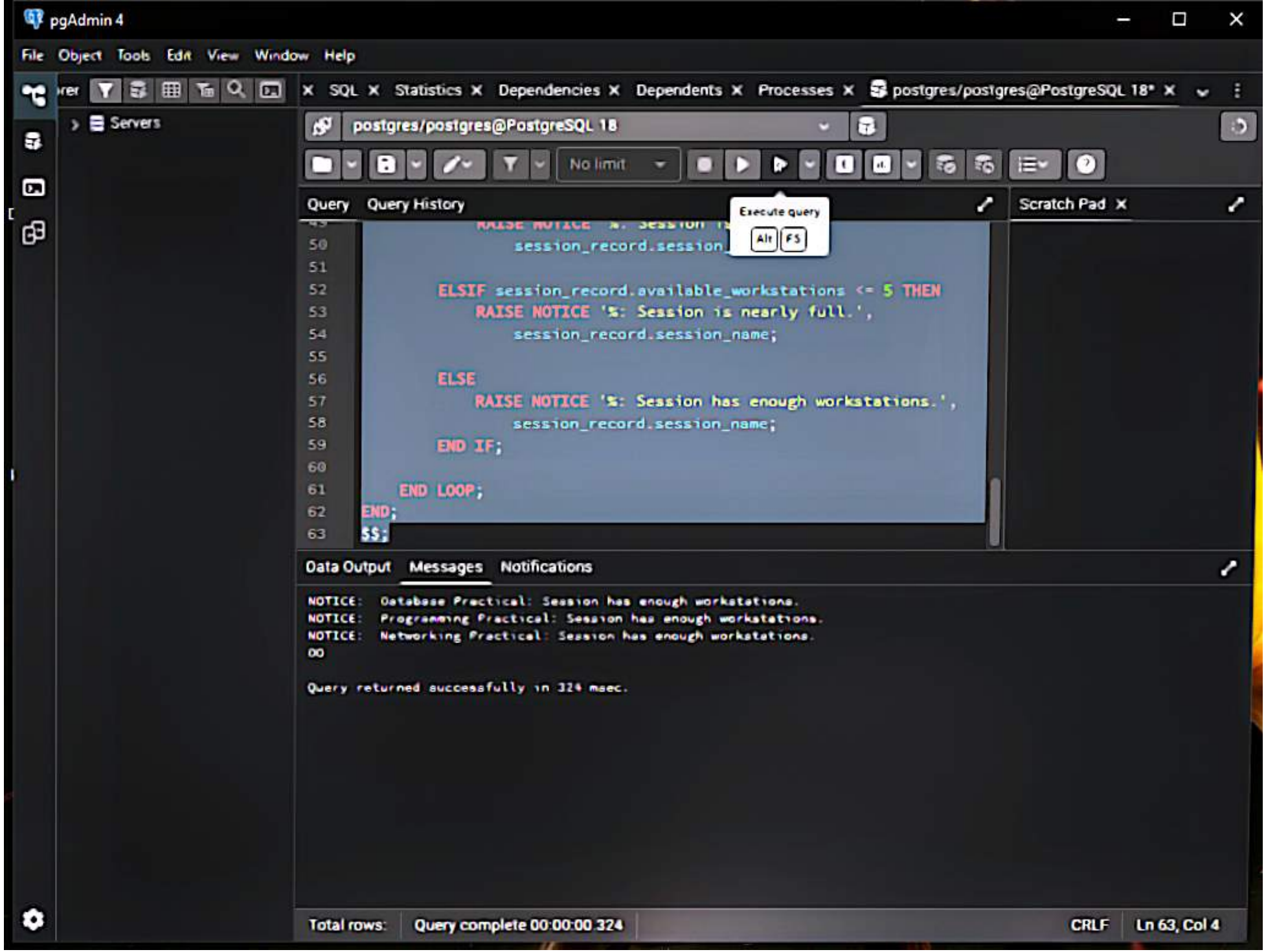
CamScanner


pgAdmin 4
File Object Tools Edit View Window Help
SQL Statistics Dependencies Dependents Processes postgres/postgres@PostgreSQL 18*
Servers
postgres/postgres@PostgreSQL 18
No limit
Query Query History
Execute query
Alt F5
Scratch Pad
reminder_number INTEGER := 1;
BEGIN
-- WHILE loop: three preparation reminders
WHILE reminder_number <= 3 LOOP
RAISE NOTICE 'Session Preparation Reminder %', reminder_n
reminder_number := reminder_number + 1;
END LOOP;
-- Numeric FOR loop: three workstation checks
FOR workstation_check IN 1..3 LOOP
RAISE NOTICE 'Workstation Check %', workstation_check;
END LOOP;
END;
$$;
Data Output Messages Notifications
NOTICE: Session Preparation Reminder 1
NOTICE: Session Preparation Reminder 2
NOTICE: Session Preparation Reminder 3
NOTICE: Workstation Check 1
NOTICE: Workstation Check 2
NOTICE: Workstation Check 3
DO
Query returned successfully in 269 msec.
Total rows:
Query complete 00:00:00.269
CRLF
Ln 107, Col 4

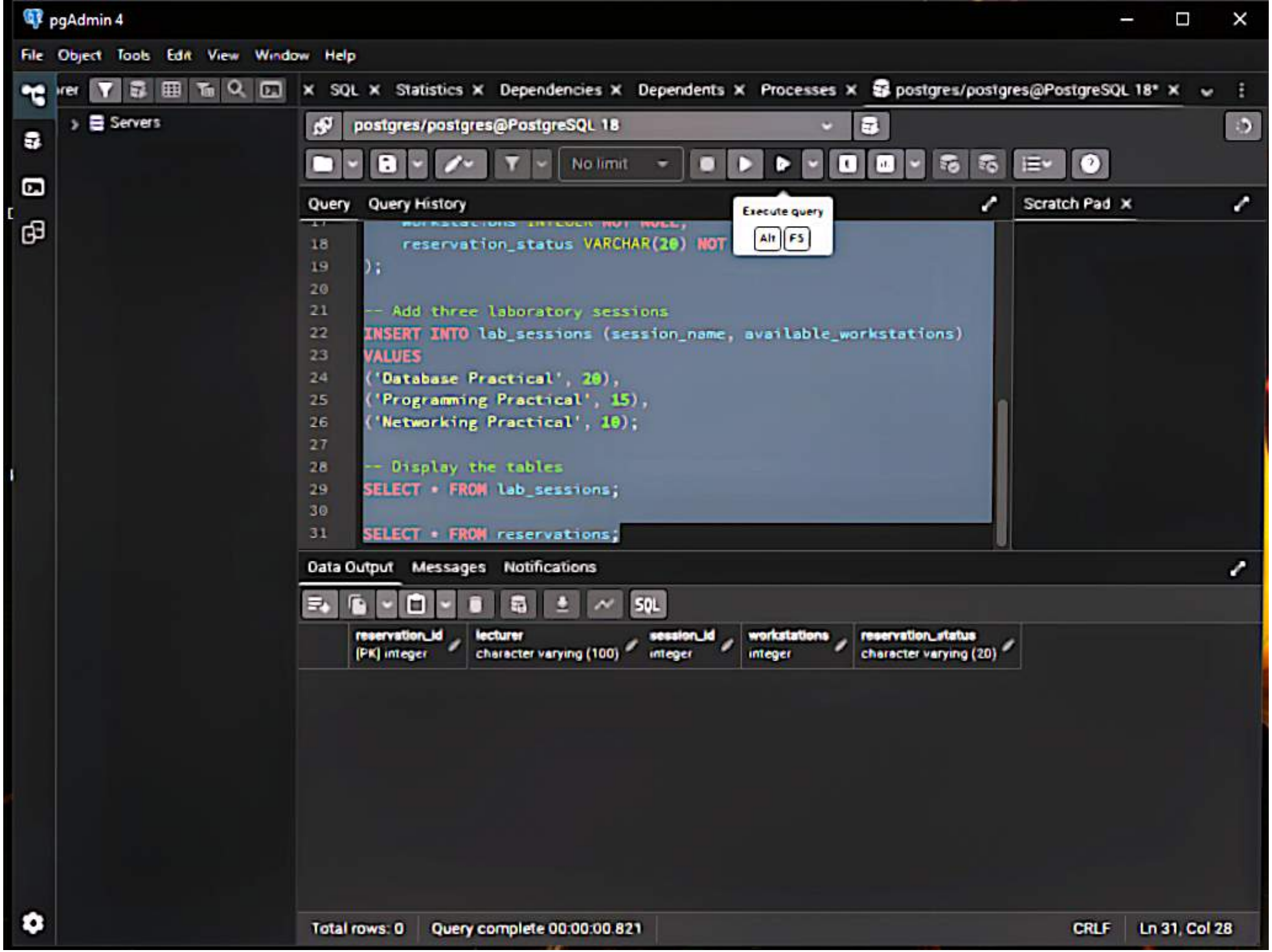
CamScanner


pgAdmin 4
File Object Tools Edit View Window Help
Servers
SQL Statistics Dependencies Dependents Processes postgres/postgres@PostgreSQL 18*
postgres/postgres@PostgreSQL 18
No limit
Query Query History
Execute query
Alt F5
Scratch Pad
74 reminder_number INTEGER := 1;
75 BEGIN
76 -- WHILE loop: three preparation reminders
77 WHILE reminder_number <= 3 LOOP
78 RAISE NOTICE 'Session Preparation Reminder %', reminder_nu
79 reminder_number := reminder_number + 1;
80 END LOOP;
81
82 -- Numeric FOR loop: three workstation checks
83 FOR workstation_check IN 1..3 LOOP
84 RAISE NOTICE 'Workstation Check %', workstation_check;
85 END LOOP;
86 END;
87 $$;
Data Output Messages Notifications
NOTICE: Session Preparation Reminder 1
NOTICE: Session Preparation Reminder 2
NOTICE: Session Preparation Reminder 3
NOTICE: Workstation Check 1
NOTICE: Workstation Check 2
NOTICE: Workstation Check 3
DO
Query returned successfully in 285 msec.
Total rows:
Query complete 00:00:00.285
CRLF
Ln 87, Col 4

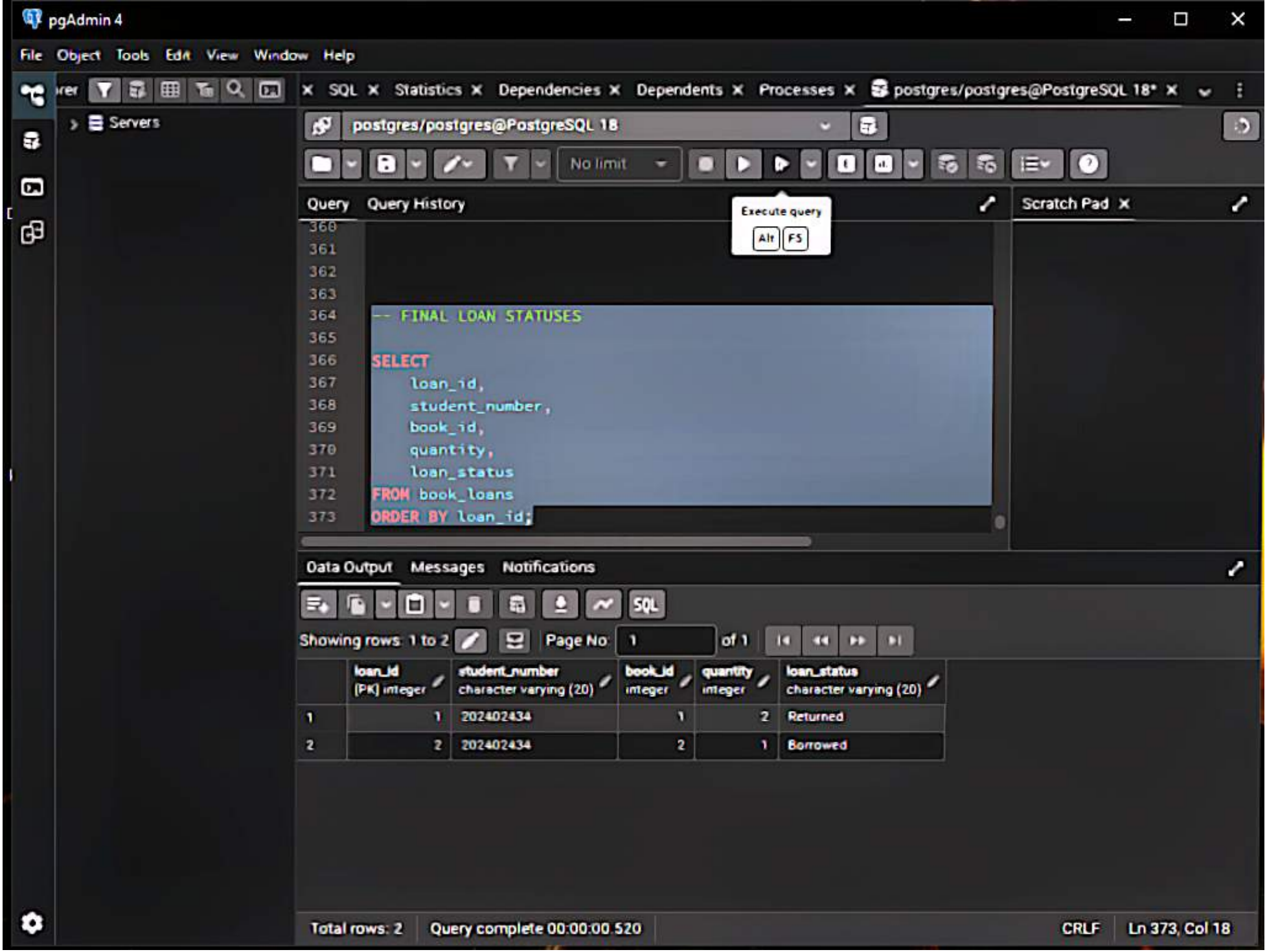
CamScanner


pgAdmin 4
File Object Tools Edit View Window Help
SQL Statistics Dependencies Dependents Processes postgres/postgres@PostgreSQL 18*
Servers
postgres/postgres@PostgreSQL 18
No limit
Query Query History
Execute query
Alt F5
Scratch Pad
50 session_record.session_
51
52 ELSIF session_record.available_workstations <= 5 THEN
53 RAISE NOTICE '%: Session is nearly full.',
54 session_record.session_name;
55
56 ELSE
57 RAISE NOTICE '%: Session has enough workstations.',
58 session_record.session_name;
59 END IF;
60
61 END LOOP;
62 END;
63 $$;
Data Output Messages Notifications
NOTICE: Database Practical: Session has enough workstations.
NOTICE: Programming Practical: Session has enough workstations.
NOTICE: Networking Practical: Session has enough workstations.
DO
Query returned successfully in 324 msec.
Total rows:
Query complete 00:00:00.324
CRLF
Ln 63, Col 4

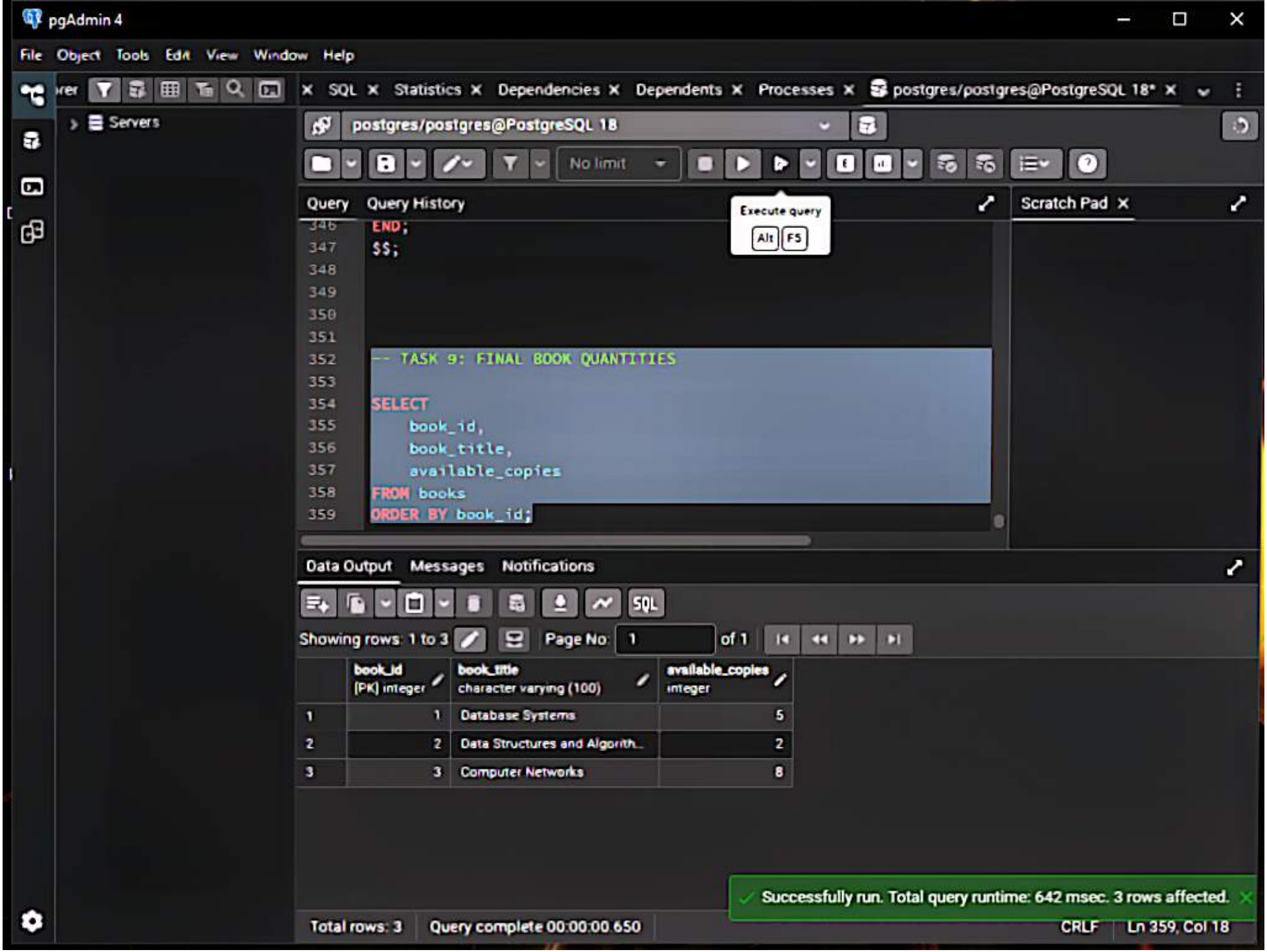
CamScanner


pgAdmin 4
File Object Tools Edit View Window Help
SQL Statistics Dependencies Dependents Processes postgres/postgres@PostgreSQL 18*
Servers
postgres/postgres@PostgreSQL 18
No limit
Query Query History
Execute query
Alt F5
Scratch Pad
reservation_status VARCHAR(20) NOT
);
-- Add three laboratory sessions
INSERT INTO lab_sessions (session_name, available_workstations)
VALUES
('Database Practical', 20),
('Programming Practical', 15),
('Networking Practical', 10);
-- Display the tables
SELECT * FROM lab_sessions;
SELECT * FROM reservations;
Data Output Messages Notifications
SQL
reservation_id (PK) integer
lecturer character varying (100)
session_id integer
workstations integer
reservation_status character varying (20)
Total rows: 0
Query complete 00:00:00.821
CRLF
Ln 31, Col 28

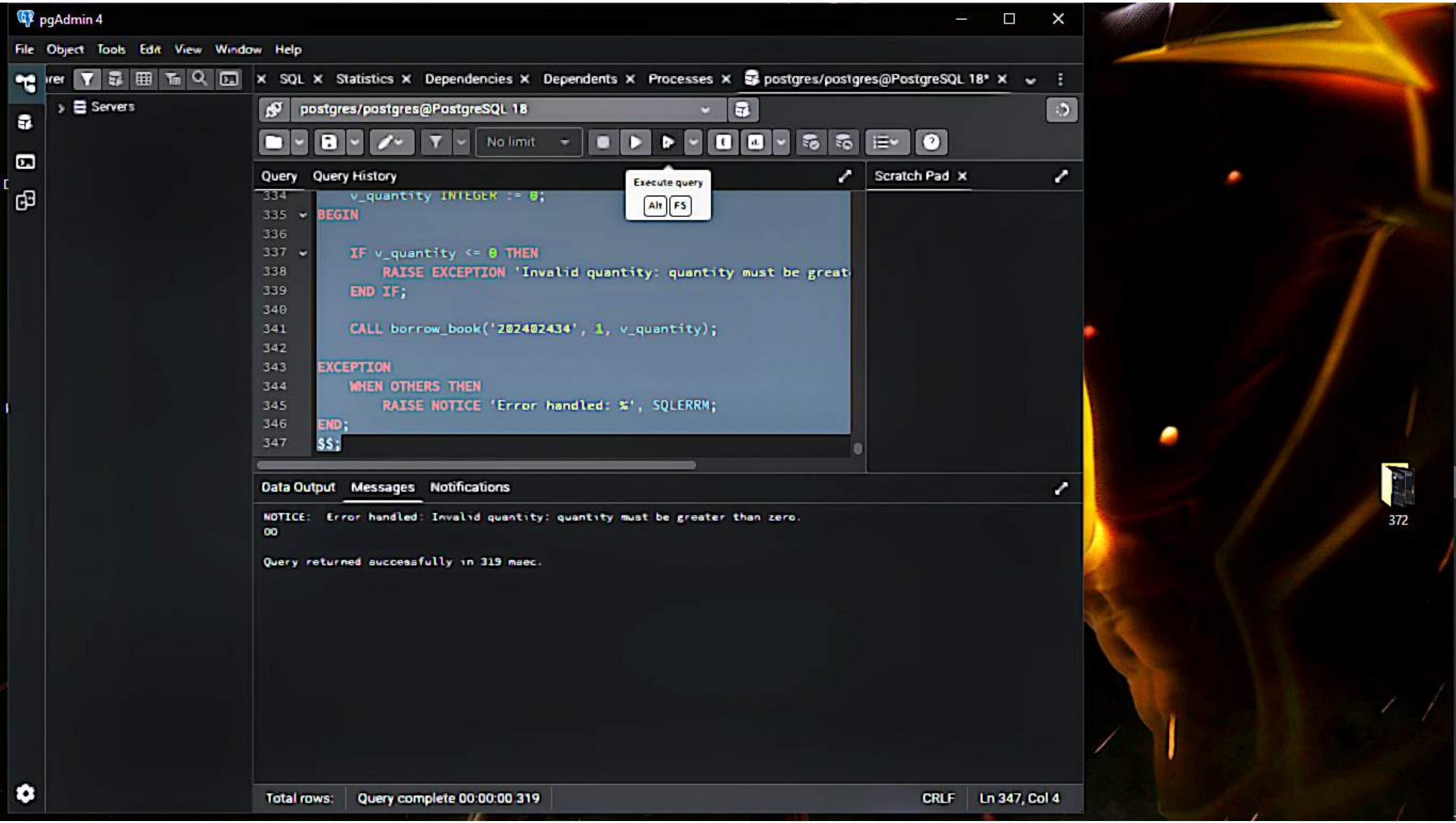
CamScanner


pgAdmin 4
File Object Tools Edit View Window Help
Servers
SQL Statistics Dependencies Dependents Processes postgres/postgres@PostgreSQL 18*
postgres/postgres@PostgreSQL 18
No limit
Query Query History
Execute query
Alt F5
Scratch Pad
360
361
362
363
364 -- FINAL LOAN STATUSES
365
366 SELECT
367 loan_id,
368 student_number,
369 book_id,
370 quantity,
371 loan_status
372 FROM book_loans
373 ORDER BY loan_id;
Data Output Messages Notifications
SQL
Showing rows: 1 to 2 Page No: 1 of 1
loan_id (PK) integer
student_number character varying (20)
book_id integer
quantity integer
loan_status character varying (20)
1 1 202402434 1 2 Returned
2 2 202402434 2 1 Borrowed
Total rows: 2 Query complete 00:00:00.520 CRLF Ln 373, Col 18

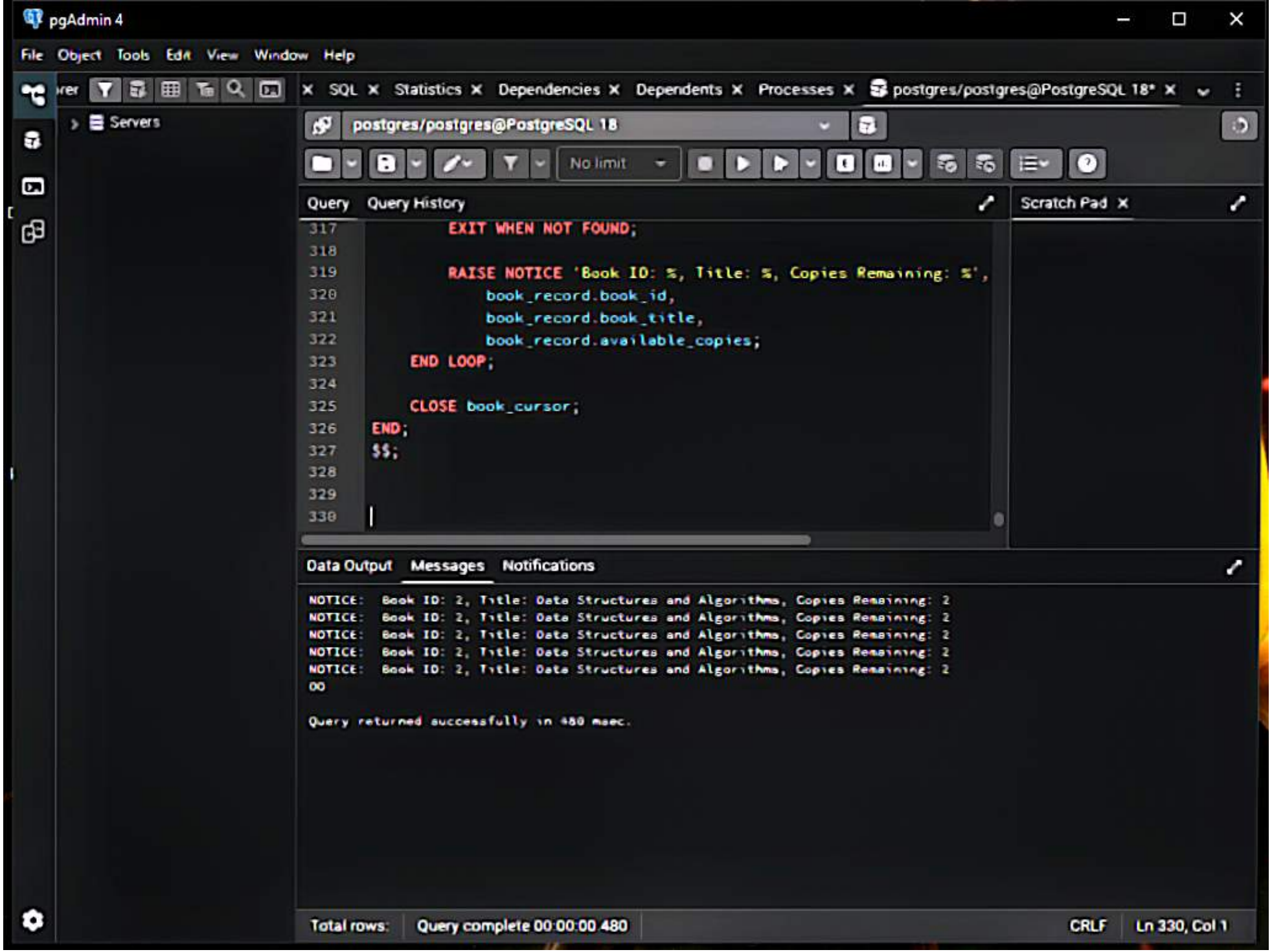
CamScanner


pgAdmin 4
File Object Tools Edit View Window Help
SQL Statistics Dependencies Dependents Processes postgres/postgres@PostgreSQL 18*
Servers
postgres/postgres@PostgreSQL 18
No limit
Query Query History
Execute query
Alt F5
Scratch Pad
END;
$$;
-- TASK 9: FINAL BOOK QUANTITIES
SELECT
book_id,
book_title,
available_copies
FROM books
ORDER BY book_id;
Data Output Messages Notifications
SQL
Showing rows: 1 to 3
Page No: 1 of 1
book_id (PK) integer
book_title character varying (100)
available_copies integer
1 1 Database Systems 5
2 2 Data Structures and Algorith... 2
3 3 Computer Networks 8
Successfully run. Total query runtime: 642 msec. 3 rows affected.
Total rows: 3
Query complete 00:00:00.650
CRLF
Ln 359, Col 18

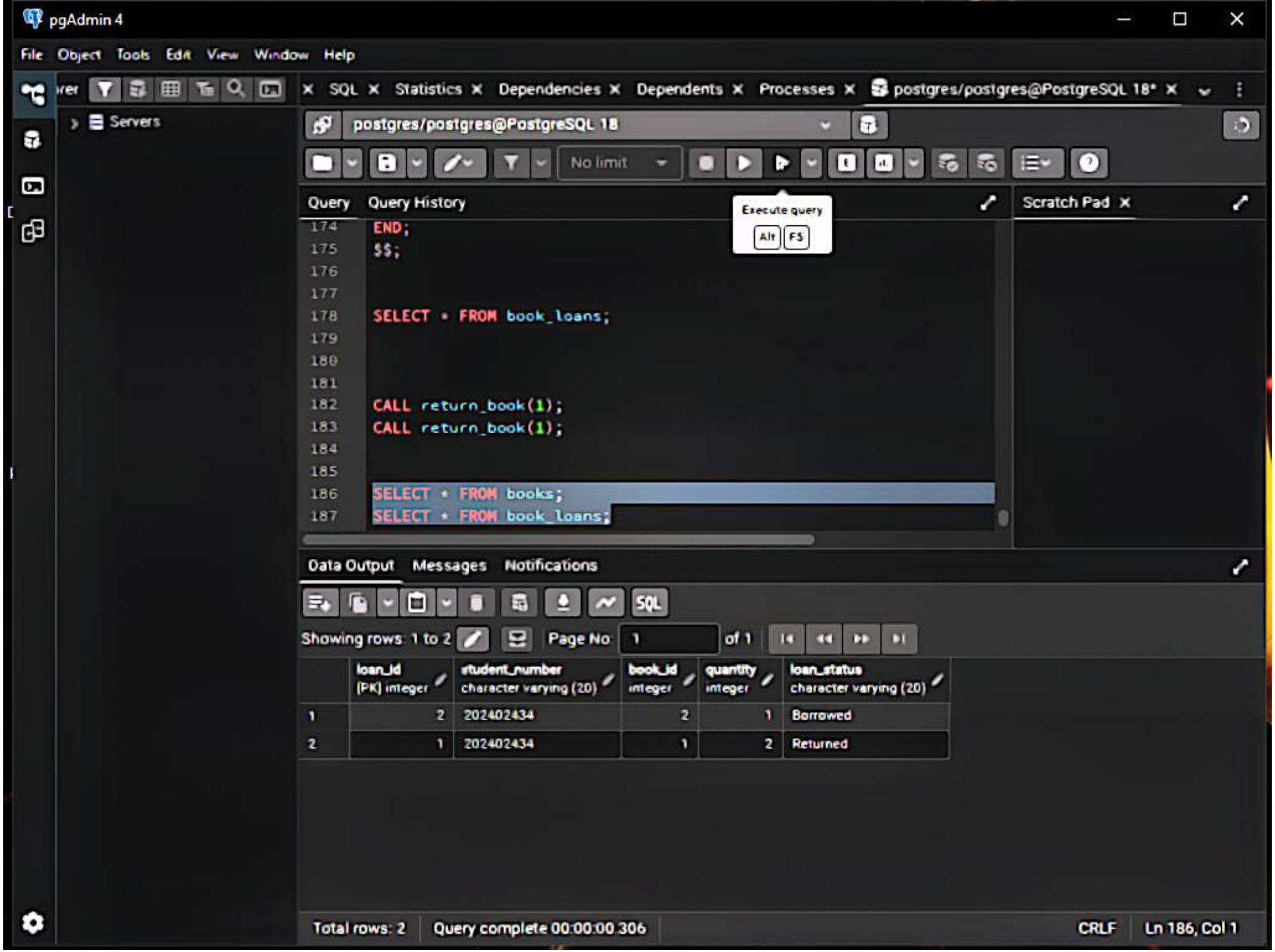
CamScanner


pgAdmin 4
File Object Tools Edit View Window Help
SQL Statistics Dependencies Dependents Processes postgres/postgres@PostgreSQL 18*
Servers
postgres/postgres@PostgreSQL 18
No limit
Query Query History
Execute query
Alt F5
Scratch Pad
v_quantity INTEGER := 0;
BEGIN
IF v_quantity <= 0 THEN
RAISE EXCEPTION 'Invalid quantity: quantity must be great
END IF;
CALL borrow_book('202402434', 1, v_quantity);
EXCEPTION
WHEN OTHERS THEN
RAISE NOTICE 'Error handled: %', SQLERRM;
END;
$$;
Data Output Messages Notifications
NOTICE: Error handled: Invalid quantity: quantity must be greater than zero.
DO
Query returned successfully in 319 msec.
Total rows:
Query complete 00:00:00.319
CRLF
Ln 347, Col 4
372

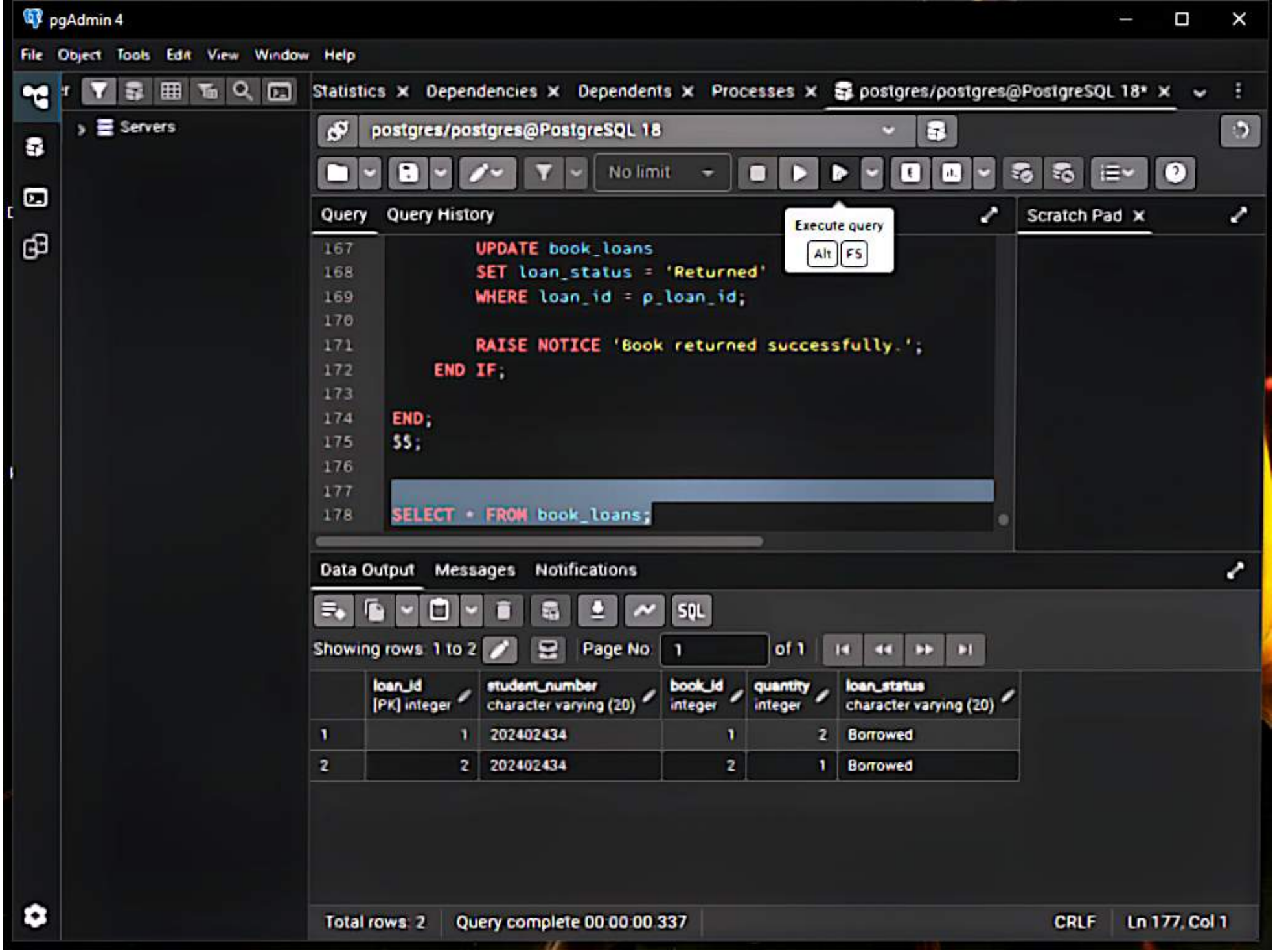
CamScanner


pgAdmin 4
File Object Tools Edit View Window Help
SQL Statistics Dependencies Dependents Processes postgres/postgres@PostgreSQL 18*
Servers
postgres/postgres@PostgreSQL 18
No limit
Query Query History
Scratch Pad
317 EXIT WHEN NOT FOUND;
318
319 RAISE NOTICE 'Book ID: %, Title: %, Copies Remaining: %',
320 book_record.book_id,
321 book_record.book_title,
322 book_record.available_copies;
323 END LOOP;
324
325 CLOSE book_cursor;
326 END;
327 $$;
328
329
330
Data Output Messages Notifications
NOTICE: Book ID: 2, Title: Data Structures and Algorithms, Copies Remaining: 2
NOTICE: Book ID: 2, Title: Data Structures and Algorithms, Copies Remaining: 2
NOTICE: Book ID: 2, Title: Data Structures and Algorithms, Copies Remaining: 2
NOTICE: Book ID: 2, Title: Data Structures and Algorithms, Copies Remaining: 2
NOTICE: Book ID: 2, Title: Data Structures and Algorithms, Copies Remaining: 2
OO
Query returned successfully in 480 msec.
Total rows:
Query complete 00:00:00.480
CRLF
Ln 330, Col 1

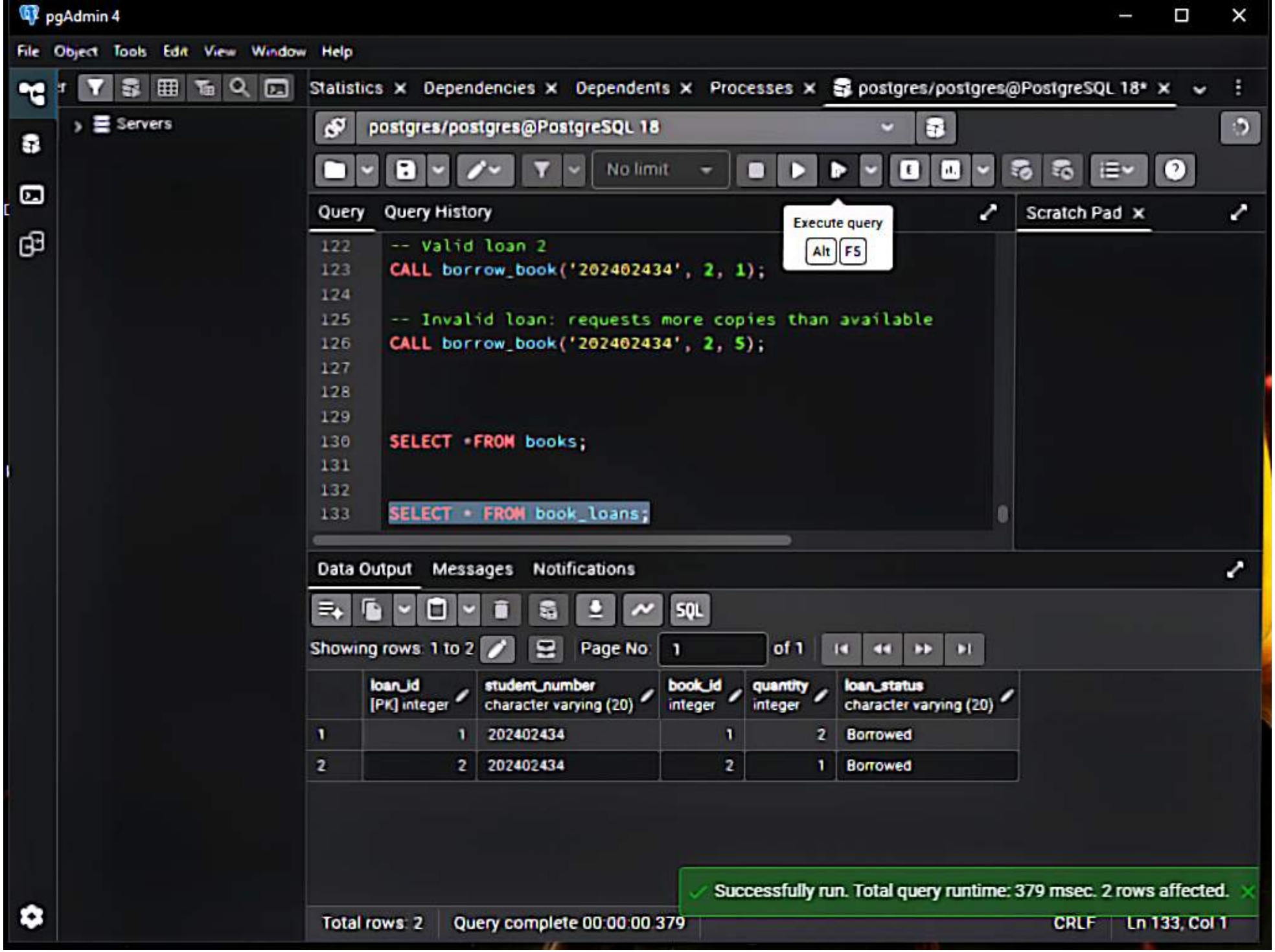
CamScanner


pgAdmin 4
File Object Tools Edit View Window Help
SQL Statistics Dependencies Dependents Processes postgres/postgres@PostgreSQL 18*
Servers
postgres/postgres@PostgreSQL 18
No limit
Query Query History
Execute query
Alt F5
Scratch Pad
174 END;
175 $$;
176
177
178 SELECT * FROM book_loans;
179
180
181
182 CALL return_book(1);
183 CALL return_book(1);
184
185
186 SELECT * FROM books;
187 SELECT * FROM book_loans;
Data Output Messages Notifications
SQL
Showing rows: 1 to 2 Page No: 1 of 1
loan_id (PK) integer
student_number character varying (20)
book_id integer
quantity integer
loan_status character varying (20)
1 2 202402434 2 1 Borrowed
2 1 202402434 1 2 Returned
Total rows: 2 Query complete 00:00:00.306 CRLF Ln 186, Col 1

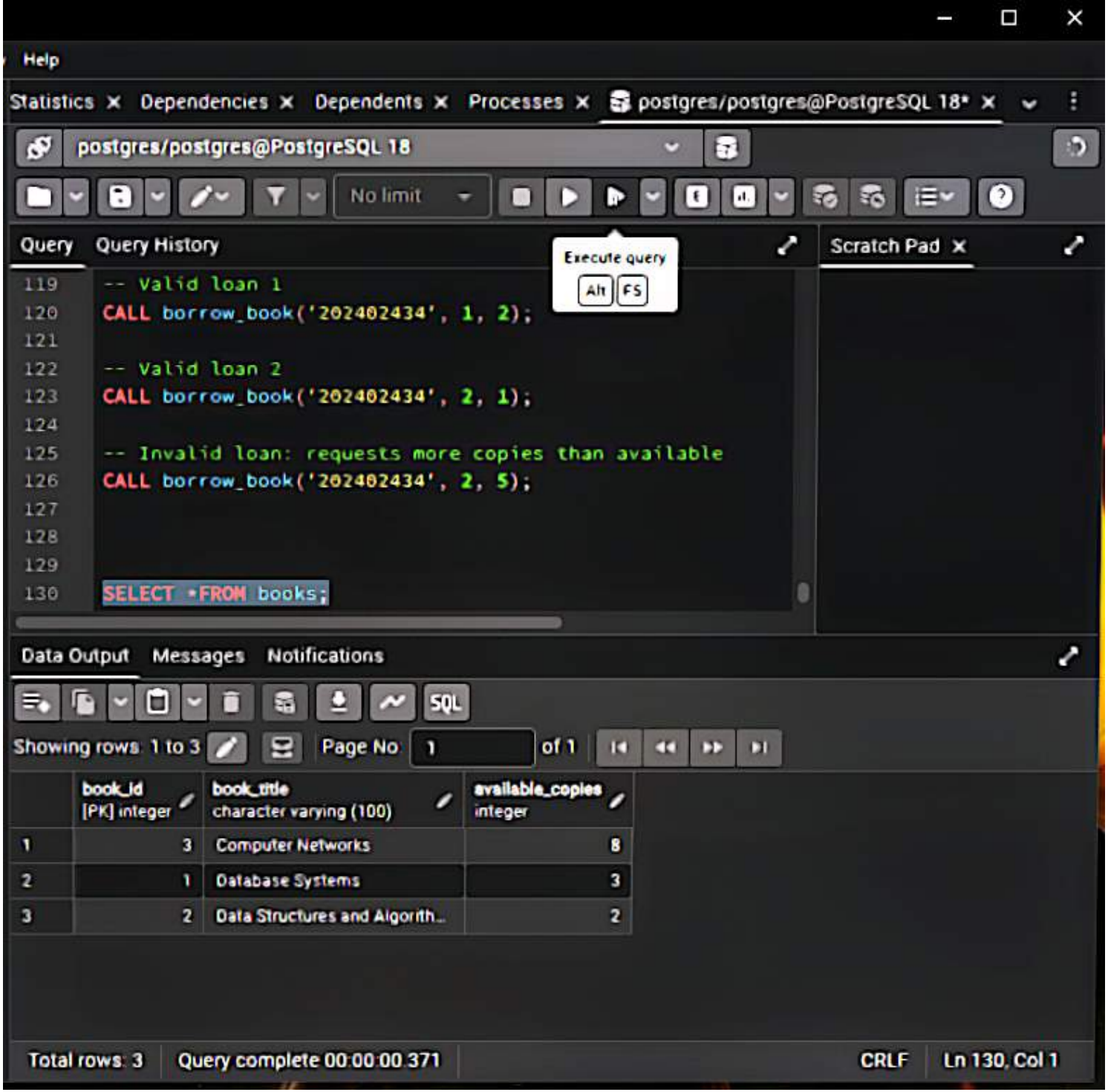
CamScanner


pgAdmin 4
File Object Tools Edit View Window Help
Statistics
Dependencies
Dependents
Processes
postgres/postgres@PostgreSQL 18*
Servers
postgres/postgres@PostgreSQL 18
No limit
Query
Query History
Execute query
Alt F5
Scratch Pad
167 UPDATE book_loans
168 SET loan_status = 'Returned'
169 WHERE loan_id = p_loan_id;
170
171 RAISE NOTICE 'Book returned successfully.';
172 END IF;
173
174 END;
175 $$;
176
177
178 SELECT * FROM book_loans;
Data Output
Messages
Notifications
SQL
Showing rows: 1 to 2
Page No 1 of 1
loan_id [PK] integer
student_number character varying (20)
book_id integer
quantity integer
loan_status character varying (20)
1 1 202402434 1 2 Borrowed
2 2 202402434 2 1 Borrowed
Total rows: 2
Query complete 00:00:00.337
CRLF
Ln 177, Col 1

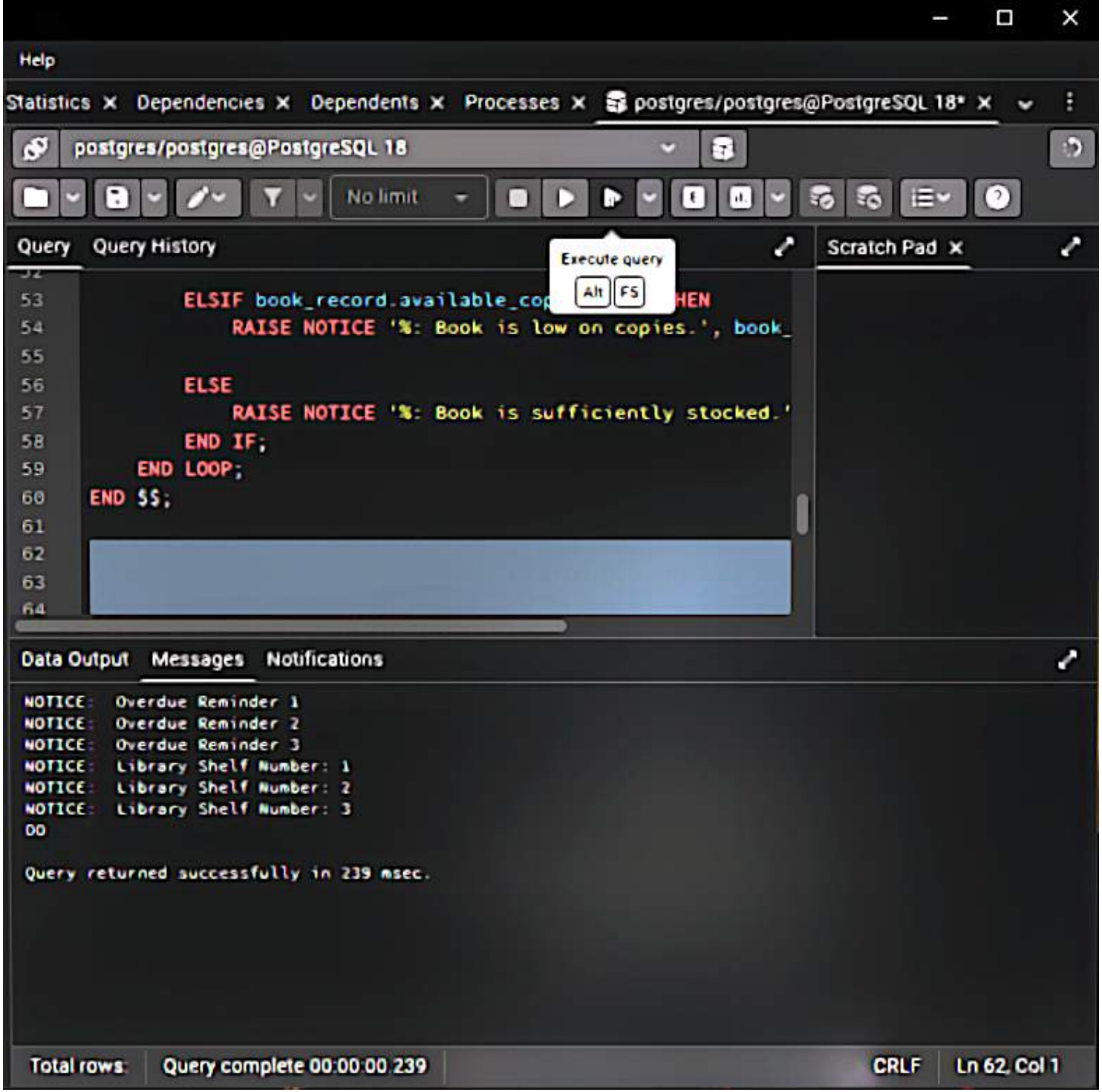
CamScanner


pgAdmin 4
File Object Tools Edit View Window Help
Statistics
Dependencies
Dependents
Processes
postgres/postgres@PostgreSQL 18*
Servers
postgres/postgres@PostgreSQL 18
No limit
Query
Query History
Execute query
Alt F5
Scratch Pad
122 -- Valid loan 2
123 CALL borrow_book('202402434', 2, 1);
124
125 -- Invalid loan: requests more copies than available
126 CALL borrow_book('202402434', 2, 5);
127
128
129
130 SELECT *FROM books;
131
132
133 SELECT * FROM book_loans;
Data Output
Messages
Notifications
SQL
Showing rows: 1 to 2
Page No 1 of 1
loan_id [PK] integer
student_number character varying (20)
book_id integer
quantity integer
loan_status character varying (20)
1 1 202402434 1 2 Borrowed
2 2 202402434 2 1 Borrowed
Successfully run. Total query runtime: 379 msec. 2 rows affected.
Total rows: 2
Query complete 00:00:00.379
CRLF
Ln 133, Col 1

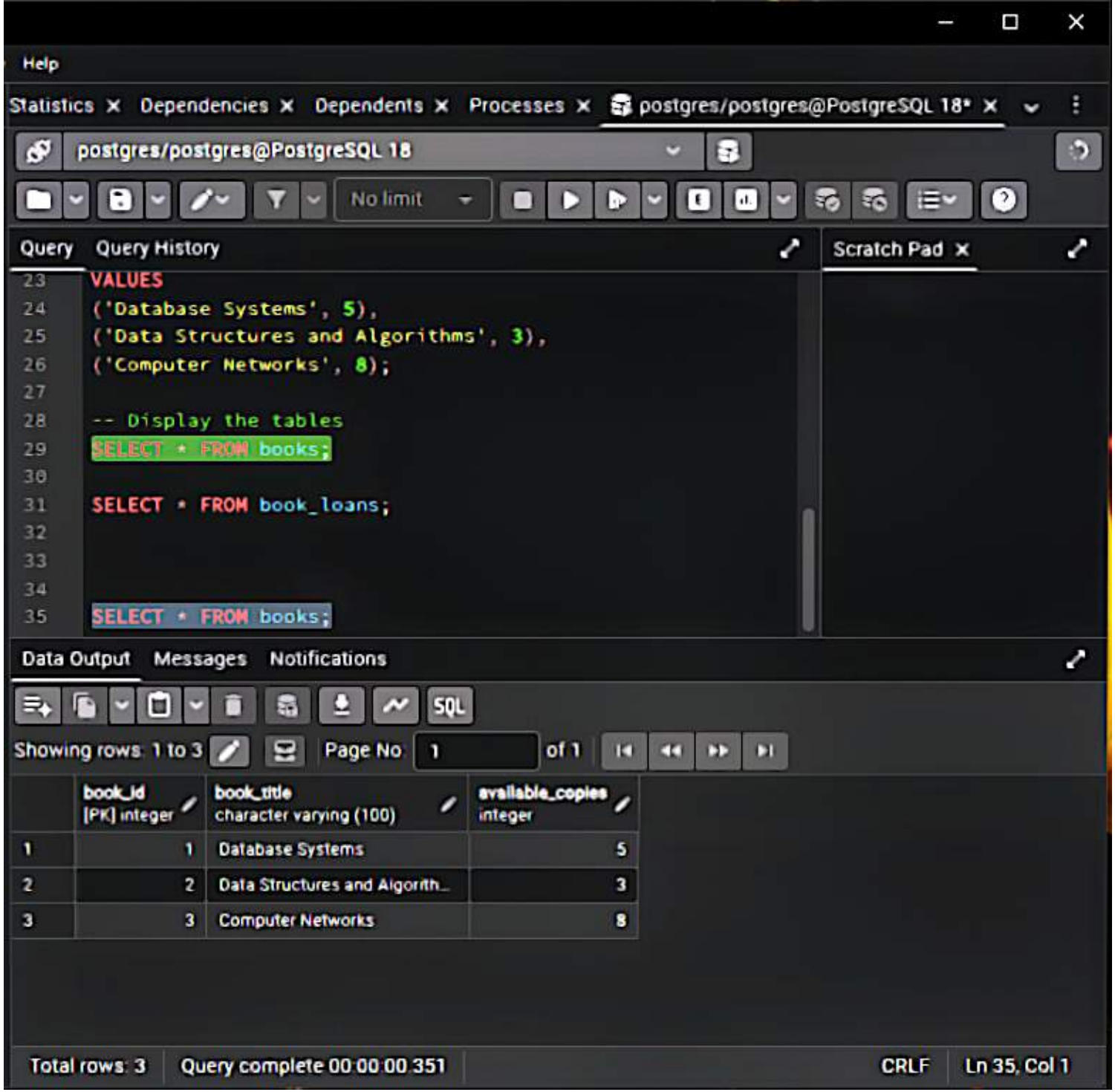
CamScanner


Help
Statistics
Dependencies
Dependents
Processes
postgres/postgres@PostgreSQL 18*
postgres/postgres@PostgreSQL 18
No limit
Query
Query History
Execute query
Alt F5
Scratch Pad
119 -- Valid loan 1
120 CALL borrow_book('202402434', 1, 2);
121
122 -- Valid loan 2
123 CALL borrow_book('202402434', 2, 1);
124
125 -- Invalid loan: requests more copies than available
126 CALL borrow_book('202402434', 2, 5);
127
128
129
130 SELECT *FROM books;
Data Output
Messages
Notifications
SQL
Showing rows: 1 to 3
Page No 1 of 1
book_id [PK] integer
book_title character varying (100)
available_copies integer
1 3 Computer Networks 8
2 1 Database Systems 3
3 2 Data Structures and Algorith... 2
Total rows: 3
Query complete 00:00:00.371
CRLF
Ln 130, Col 1

CamScanner


Help
Statistics
Dependencies
Dependents
Processes
postgres/postgres@PostgreSQL 18*
postgres/postgres@PostgreSQL 18
No limit
Query
Query History
Execute query
Alt F5
Scratch Pad
ELSIF book_record.available_cop HEN
RAISE NOTICE '%: Book is low on copies.', book_
ELSE
RAISE NOTICE '%: Book is sufficiently stocked.'
END IF;
END LOOP;
END $$;
Data Output
Messages
Notifications
NOTICE: Overdue Reminder 1
NOTICE: Overdue Reminder 2
NOTICE: Overdue Reminder 3
NOTICE: Library Shelf Number: 1
NOTICE: Library Shelf Number: 2
NOTICE: Library Shelf Number: 3
DO
Query returned successfully in 239 msec.
Total rows:
Query complete 00:00:00.239
CRLF
Ln 62, Col 1

CamScanner


Help
Statistics
Dependencies
Dependents
Processes
postgres/postgres@PostgreSQL 18*
postgres/postgres@PostgreSQL 18
No limit
Query
Query History
Scratch Pad
VALUES
('Database Systems', 5),
('Data Structures and Algorithms', 3),
('Computer Networks', 8);
-- Display the tables
SELECT * FROM books;
SELECT * FROM book_loans;
SELECT * FROM books;
Data Output
Messages
Notifications
SQL
Showing rows: 1 to 3
Page No
1
of 1
book_id [PK] integer
book_title character varying (100)
available_copies integer
1 1 Database Systems 5
2 2 Data Structures and Algorith... 3
3 3 Computer Networks 8
Total rows: 3
Query complete 00:00:00.351
CRLF
Ln 35, Col 1

CamScanner